Mohsin Ibna Hossain
AIUB, DM Note,

#Final Term.

lecture no: 13

Relation and their properties

⇒ The most direct way to express a relationship between elements of two sets is to use ordered pairs made up of two related elements. For this reason, sets of ordered pairs are called binary relations.

Relations:

⇒ Relations are derived from Cartesian product.

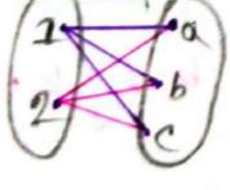
Ex:

$A = \{1, 2\}$

$B = \{a, b, c\}$

$A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c)\}$

Graphically:



⇒ Let A and B be two non-empty sets. Then a Relation R from A to B is a subset of $A \times B$

denote by $R \subseteq A \times B$

Problem: 1 $A = \{1, 2\}$ and $B = \{1, 2, 3\}$ then $R_1 = \{(1,1), (1,2), (2,1), (2,2)\}$ Can it be called a relation?

⇒ Is given:

$A = \{1, 2\}$

$B = \{1, 2, 3\}$

$A \times B = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3)\}$

Since, Every element of R_1 present in $A \times B$,

So, R_1 is a Relation.

Binary Relation:—

⇒ Let, A and B be two sets. A binary Relation R from A to B is a subset of $A \times B$. Denote by $R \subseteq A \times B$.

1. We use the notation aRb to denote that $(a,b) \in R$

2. when (a,b) belongs to R,
   a is said to be related to b by R.

3. $a \not R b$ means a is not related to b by R. denote that $(a,b) \notin R$

Problem:— Let, $A = \{0,1,2\}$ and $B = \{a,b\}$.

Then $R' = \{(0,a), (0,b), (1,a), (1,b), (2,b)\}$ Can it be called a relation.

⇒ In given:—

$A = \{0,1,2\}$
$B = \{a,b\}$

$R' = \{(0,a), (0,b), (1,a), (1,b), (2,b)\}$

$A \times B = \{(0,a), (0,b), (1,b), (1,a), (2,a), (2,b)\}$

Since, Every element of R' is present in A×B

So: yes, R' is a Relation.

Relations can be represented Graphically or using Table:—

Graphically:—

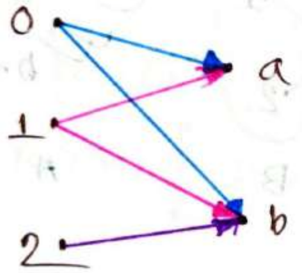


Table:—

R	a	b
0	×	×
1	×	×
2		×

Functions as Relations:—

⇒ Let, A and B two sets. A relation 'R' from A to B is called function if each element of A, we can assign unique element of B.

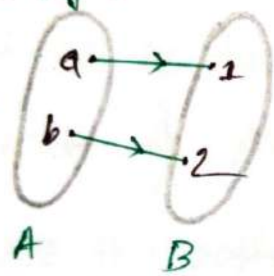
Ex:—

$A = \{a, b\}, B = \{1, 2\}$

$A \times B = \{(a,1), (a,2), (b,1), (b,2)\}$

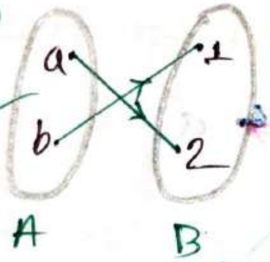
Graphically:—

(1.) ✓

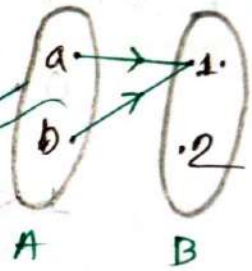


It's a relation and a function

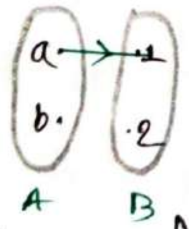
(2.) ✓



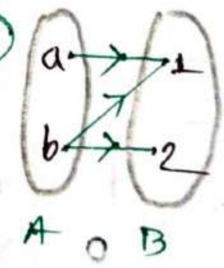
(3.) ✓



✗ (4.)



✗ (5.)



It's not a function but it's a relation.

Function Vs Relations:—

(1) A relation can be used to express a one to many relationship between the elements of the sets A and B. Where an element of A may be related to more than one element of B.

(2) A function represents a relation where exactly one element of B in related to each element of A.

(3) Relations are more general than functions. A function in related to where exactly one element of B in related to each element of A.

Relations on a Set

Relation from a set to itself:—

⇒ A Relation on a set A in a relation from A to A. In other words, a relation on a set in a subset of $A \times A$.

#problem 42— Let A be the set $\{1,2,3,4\}$. Which ordered Pairs are in the relation $R = \{(a,b) \mid a \text{ divides } b\}$?

In given:—

$$A = \{1,2,3,4\}$$

$$\therefore A \times A = \left\{ \begin{matrix} (1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4) \\ (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4) \end{matrix} \right\}$$

$\therefore R = \{(1,1),(1,2),(1,3),(1,4),(2,2),(2,4),(3,3),(4,4)\}$

Problem 5:—

$\Rightarrow$ Consider these relations on the set of integers.

$R_1 = \{(a,b) \mid a \leq b\}$,

$R_2 = \{(a,b) \mid a > b\}$,

$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\}$,

$R_4 = \{(a,b) \mid a = b\}$

$R_5 = \{(a,b) \mid a = b+1\}$

$R_6 = \{(a,b) \mid a+b \leq 3\}$.

Which of these relations contain each of the pairs $(1,1), (1,2), (2,1), (1,-1), (2,2)$?

$\Rightarrow$ Checking the conditions that define each relation, we see that the pair

1. $(1,1)$ is in R_1, R_3, R_4 and R_6
2. $(1,2)$ is in R_1, R_6
3. $(2,1)$ is in R_2, R_5, R_6
4. $(1,-1)$ is in R_2, R_3, R_6
5. $(2,2)$ is in R_1, R_3, R_4,

Properties of Relations:-

⇒ There are several properties that are used to classify relation on a set.

here is the most important properties:-

1. Reflexive
2. Symmetric
3. antisymmetric
4. Transitive.

Reflexive Relation:-

⇒ A relation R on a set A is called reflexive if $(a,a) \in R$ for every element $a \in A$.

⇒ Using quantifiers, a relation on the set A is reflexive if $\forall a((a,a) \in R)$, where Domain is the set of All Elements in A.

1. In a reflexive relation, every element is related to itself.

Problem:— Let $A = \{1,2,3\}$ and the relation $R = \{(1,1),(1,2),(2,1),(2,2),(3,3)\}$ Can it be called a reflexive relation.

⇒ In given:—

$A = \{1,2,3\}$

$R = \{(1,1),(1,2),(2,1),(2,2),(3,3)\}$

We know:—

⇒ A relation R on a set A is called a reflexive if $(a,a) \in R$ for every element $a \in A$

So All the pairs of the form (a,a), namely $(1,1),(2,2),(3,3)$

Then:– The relation R is a reflexive relation because it contain All the pairs of (a,a)

Problem 7:— Let $A = \{1,2,3,4\}$. Determine whether the following relations are Reflexive or not.

1. $R_1 = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)\}$
2. $R_2 = \{(1,1),(1,2),(2,1)\}$
3. $R_3 = \{(1,1),(1,2),(1,4),(2,1),(2,2),(3,3),(4,4)\}$
4. $R_4 = \{(2,1),(3,1),(3,2),(4,1),(4,3)\}$
5. $R_5 = \{(1,1),(2,2),(3,3),(4,4)\}$
6. $R_6 = \{(3,4)\}$.

⇒ We know:-

⇒ A relation R on a set A is called a reflexive if $(a,a) \in R$ for every element $a \in A$.

So, All the pairs of the form (a,a) is $(1,1)(2,2)(3,3)$ $(4,4)$.

Then:- The relation R_3 and R_5 is a reflexive relation because they contain All the pairs of the form(a,a).

problem: 8:-

⇒ Determining whether the following relations are reflexive or not. on the set of integers.

1. $R_1 = \{(a,b) \mid a \leq b\}$
2. $R_2 = \{(a,b) \mid a > b\}$
3. $R_3 = \{(a,b) \mid a = b \text{ or } a = -b\}$
4. $R_4 = \{(a,b) \mid a = b\}$
5. $R_5 = \{(a,b) \mid a = b+1\}$
6. $R_6 = \{(a,b) \mid a+b \leq 3\}$

⇒ Checking the condition that define reflexive or not reflexive relation. We can see that.

(1.) R_1, R_3, R_4 is a reflexive relation.

→ because they contain all the pairs of the from (a, a).

(2.) R_2 is not a reflexive relation.

→ because $(1 \not R 1)$ it not satisfy the condition.

(3.) R_5 is not a reflexive relation.

→ because $(1 \neq 1+1)$

(4.) R_6 is not a reflexive relation.

→ because $(1+1 \not\leq 3)$

Problem of:

⇒ In the 'divides' relation on the set of positive integer reflexive?

⇒ Let:

The set of positive integer $A = \{1, 2, 3, 4, \ldots\}$

Now:
$(1|1) = 1$
$(2|2) = 1$
$(3|3) = 1 \ldots$

So, yes, the devides relation on the set of positive integer is a reflexive relation.

Problem: 10:

⇒ In the devides relation on the set of integers reflexive?

⇒ Let: The set of integer $A = \{..., -1, 0, 1, ...\}$

Now: $(0,0)$ = (0 does not divide 0)

So, The devides relation on the set of integer is not a reflexive relation.

Symmetric Relation:

⇒ A relation R on a set A is called symmetric if $(b,a) \in R$ whenever $(a,b) \in R$ for all $a, b \in A$

1. if $(x,y) \in R$ then $(y,x) \in R$, $\forall (x,y) \in A$
2. A null set also a symmetric Relation.

Problem: Let, $A = \{1, 2, 3, 4\}$ and the relation $R = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,3), (1,4)\}$ can it be called a Symmetric Relation?

⇒ In given:—

$A = \{1, 2, 3, 4\}$

$R = \{(1,1), (1,2); (2,1); (2,2), (3,4), (4,1), (4,3), (1,4)\}$

We know:—

⇒ A relation R on a set A is called symmetric if $(a,b) \in R$ then $(b,a) \in R$ for $\forall (a,b) \in A$.

Since:— The element of relation R is satisfy the condition of symmetric Relation.

So The Relation R is a symmetric Relation.

Antisymmetric Relation:—

⇒ A relation R on a set A such that for all $(a,b) \in A$ if $(a,b) \in R$ and $(b,a) \in R$, than $a=b$ is called Antisymmetric.

1. If $(a,b) \in R$ then $(b,a) \in R$ only if $x=y$ for $\forall (a,b) \in A$
2. A null set/empty set also Antisymmetric
3. R is antisymmetric if whenever $a=b$ then $a \not R b$ or $b \not R a$.
4. R is not antisymmetric if we have a and b in A, $a \neq b$ and both aRb or bRa

problem :- 16

Let $A = \{0, 1, 2\}$ and the Relations :

(1) $R_1 = \{(1,1), (2,2)\}$

(2) $R_2 = \{(0,1), (1,2), (2,1)\}$

(3) $R_3 = \{(0,1), (1,1), (2,1), (2,2)\}$

(4) $R_4 = \{(0,1), (1,1), (1,0), (2,2)\}$

Determine whether the following relations are Symmetric or Antisymmetric or both or not.

=> We know :-

=> A relation is a symmetric if $(a,b) \in R$ then $(b,a) \in R$ for $\forall (a,b) \in A$

=> A relation is Antisymmetric if $(a,b) \in R$ then $(b,a) \in R$ only if $x = y$ for $\forall (a,b) \in A$

So, (1) R_1 is both symmetric and Antisymmetric

(2) R_2 is not a symmetric and not Antisymmetric

(3) R_3 is not symmetric but R_3 is Antisymmetric

(4) R_4 is symmetric but not Antisymmetric

Problem 10:— Let, $A = \{1,2,3,4\}$ and the relations on

1. $R_1 = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)\}$
2. $R_2 = \{(1,1),(1,2),(2,1)\}$
3. $R_3 = \{(1,1),(1,2),(1,4),(2,1),(2,2),(3,3),(4,1),(4,4)\}$
4. $R_4 = \{(2,1),(3,1),(3,2),(4,1),(4,2),(4,3)\}$
5. $R_5 = \{(1,1),(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,3),(3,4),(4,4)\}$
6. $R_6 = \{(3,4)\}$.

⇒ We know:—

⇒ A relation in symmetric if $(a,b) \in R$ then $(b,a) \in R$ for $\forall (a,b) \in A$

⇒ A relation is Antisymmetric if $(a,b) \in R$ then $(b,a) \in R$ only if $x = y$ for $\forall (a,b) \in A$

So,

1. R_1 is not a symmetric and not Antisymmetric
2. R_2 and R_3 in a symmetric
3. R_4, R_5 and R_6 in anti-symmetric

Transitive Relation:-

⇒ A relation R on a set A called transitive if whenever $(a,b) \in R$ and $(b,c) \in R$, then $(a,c) \in R$

(1) if $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$

(2) A empty set always a Transitive.

Problem:- let $A = \{1,2,3,4\}$ and the relations $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$ Can it be called a Transitive.

⇒ We know:-

⇒ A relation is transitive if $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$.

So,

⇒ All elements of the R relation satisfy the symmetric relation condition.

Then:- The relation R in a Transitive.

Problem 13:-

⇒ let $A = \{1, 2, 3, 4\}$ and the relations

$R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$

$R_2 = \{(1,1), (1,2), (2,1)\}$

$R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$

$R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$

$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$

$R_6 = \{(3,4)\}$

Which of the relations are transitive?

⇒ (1.) R_1 not transitive. because $(3,4)$ and $(4,3)$ belong to R_1 but $(3,1)$ does not.

(2.) R_2 not transitive. because $(2,1)$ and $(1,2)$ belong to R_2 but $(2,2)$ does not.

(3.) R_3 not transitive. because $(4,1)$ and $(1,2)$ belong to R_3 but $(4,2)$ does not.

(4.) R_4, R_5, R_6 is transitive. because that if (a,b) and (b,c) belong to this relation then (a,c) belongs to the relation. its verify.

Problem: 11:—

⇒ Is the relation $R = \{(a,a), (b,c), (c,b), (d,d)\}$ on the set $X = \{a,b,c,d\}$ is transitive?

⇒ No. It's not transitive.

Because (b,c) and (c,b) are in R, but (b,b) is not in R.

Combining Relations:—

⇒ Because relations from A to B are subset of $A \times B$ two relations form A to B can be combined in any way two sets can be combined.

⇒ Let two relation R_1 and R_2, we can combine them using basic set operations to form new relations such as $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$ and $R_2 - R_1$

Problem: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$ the relations $R_1 = \{(1,1), (2,2), (3,3)\}$ and $R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$ can be combined using basic set operations to form new relations.

⇒ In given:—

$A = \{1, 2, 3\}$

$B = \{1, 2, 3, 4\}$

$R_1 = \{(1,1), (2,2), (3,3)\}$

$R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$

$\therefore R_1 \cup R_2 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (3,3)\}$

$R_1 \cap R_2 = \{(1,1)\}$

$R_1 - R_2 = \{(2,2), (3,3)\}$

$R_2 - R_1 = \{(1,2), (1,3), (1,4)\}$

Composite of Relations:—

⇒ Let, A, B, and C be sets.

~~Let~~ R is the relation from A to B that is $R \subseteq A \times B$

S is the relation from B to C that is $S \subseteq B \times C$

1. The composition of R and S denote by $S \circ R$.
2. $R \circ S = \{(a,c) \in A \times C : \text{for some } b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$

problem 29

$\Rightarrow$ what is the composite of the Relations R and S where R is the relation from {1,2,3} to {1,2,3,4} with $R=\{(1,1),(1,4),(2,3),(3,1),(3,4)\}$ and S is the Relation form {1,2,3,4} to {0,1,2} with $S=\{(1,0),(2,0),(3,1),(3,2),(4,1)\}$?

$\Rightarrow$ Is given:

$$R=\{(1,1),(1,4),(2,3),(3,1),(3,4)\}$$
$$S=\{(1,0),(2,0),(3,1),(3,2),(4,1)\}$$

$$\therefore SoR=\{(1,0),(1,1),(2,1),(2,2),(3,0),(3,1)\}$$

Problem 30:

$\Rightarrow$ Let R be the relation $\{(1,2),(1,3),(2,3),(2,4),(3,1)\}$ and S be the relation $\{(2,1),(3,1),(3,2),(4,2)\}$ Find SoR.

$\Rightarrow$ Is given:

$$R=\{(1,2),(1,3),(2,3),(2,4),(3,1)\}$$
$$S=\{(2,1),(3,1),(3,2),(4,2)\}$$

$$\therefore SoR=\{(1,1),(1,2),(2,1),(2,2)\}$$

Lecture: 14:-

Representing Relations:-

⇒ In this section we will discuss two alternative methods of representing relations:-

1. Representing Relations using Matrices (0, 1 matrice
2. Representing Relations using Directed graph (Digraph)

Representing Relation Using Matrices:-

⇒ A relation between finite sets can be represented using a zero-one matrix.

⇒ If R is a relation from $A = \{a_1, a_2, a_3, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$, then R can be represented by the zero-one matrix $M_R = [m_{ij}]$ with

1. $m_{ij} = 1$, if $(a_i, b_j) \in R$, and
2. $m_{ij} = 0$, if $(a_i, b_j) \notin R$

☒ For creating this matrix we first need to list the elements in A and B in particular but arbitary order.

Problem 1:

⇒ Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$ let R be the relation from A to B containing (a, b) if $a \in A$, $b \in B$, and $a > b$. What in the matrix representing R.

⇒ Is given:

$$A = \{1, 2, 3\}$$
$$B = \{1, 2\}$$
$$\therefore R = \{(2,1), (3,1), (3,2)\}$$

$$\therefore \text{The matrix } M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{cc} 1 & 2 \end{array} \\ \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \end{array}$$

Problem 2:

⇒ Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$
Which ordered pairs are in the relation R represented by the matrix

$$M_R = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array}\begin{array}{c} \begin{array}{ccccc} b_1 & b_2 & b_3 & b_4 & b_5 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array} ?$$

⇒ Because R consists of those ordered pairs (a_i, b_j) with $m_{ij} = 1$, its follows that.

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_5)\}$$

Matricees of Relation on sets:—

Reflexive relation Matricees:—

① If R is a reflexive relation, All the element of the main diagonal of M_R are equal to 1.

That means:— $\boxed{M_{ii} = 1}$

Example: Let $A = \{1, 2, 3\}$

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Symmetric Relation Matrix:—

① R in a symmetric Relation Matrix

(i) if and only if $\boxed{m_{ij} = m_{ji}}$

(ii) if and only if $\boxed{M_R = (M_R)^t}$

① $(M_R)^t$ = The transpose matrix of M_R

Example:— Let $A = \{1, 2, 3\}$

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \end{array}$$

Antisymmetric Matrices:—

⇒ R is a Antisymmetric Matrices

(i) if $m_{ij} = 1$ then $m_{ji} = 0$

(1) For Antisymmetry, there can never be two 1's symmetrically placed about the main diagonal.

Example:—

Let $A = \{1, 2, 3\}$

$$M_R = \begin{array}{c c} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$$

$$M_R = \begin{array}{c c} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$

Problem 3:—

⇒ Suppose that the relation R on a set is represented by the matrix.

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Is R reflexive, symmetric or antisymmetric?

⇒ we know:—

(1.) R is reflexive if all the diagonal elements are equal to 1. that means $m_{ii} = 1$

so, R is reflexive.

(2.) R is a symmetric if and only if $m_{ij} = m_{ji}$

so, R is symmetric.

(3.) R is antisymmetric if $m_{ij} = 1$ then $m_{ji} = 0$

so, R is not antisymmetric; because both $m_{1,2}$ and $m_{2,1}$ are 1

Representing Relations Using Digraphs:

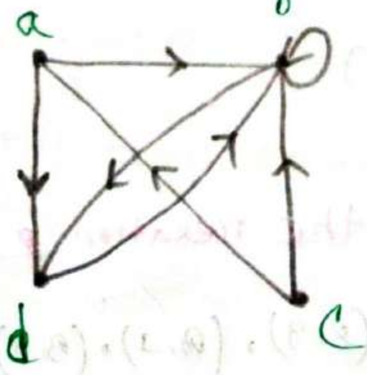
⇒ A directed graph, or digraph, consists of a set V of vertices (or nodes) together with a set E of ordered pairs of elements of V called edges. The vertex a is called the initial vertex of the edge (a,b), and the vertex b is called the terminal vertex of this edge.

① An edge of the form (a,a) is called a loop.

Problem: 7:

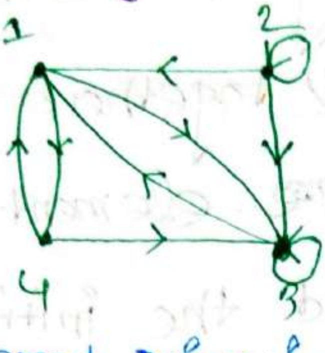
⇒ A drawing of the directed graph with vertices a, b, c, d and edge (a,b), (a,d), (b,b), (b,d), (c,a), (c,b), (d,b) is shown below.

⇒



problem:-

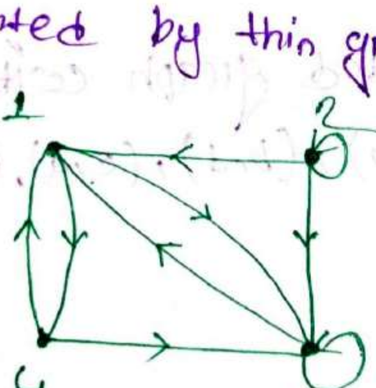
⇒ What are the ordered pairs in the relation represented by this directed graph?



⇒ The ordered pairs in the relation are
(1,3), (1,4), (2,1), (2,2), (2,3), (3,1), (3,3), (4,1), (4,3)

problem:-

⇒ What are the ordered pair in the relation R represented by this graph?



⇒ The ordered pairs in the relation R are:-
(1,3) (1,4), (2,1), (2,2), (2,3), (3,1), (3,3), (4,1), (4,3)

Determining which properties A Relation has form its Digraph:—

Reflexivity:—

⇒ A loop must be present at all vertices in the graph.

Ex:- ① ✓

② ✗

Symmetry:—

⇒ If (a, b) is an edge, then so is (b, a).

Ex:- ① ✓

② ✗

Antisymmetry:—

⇒ Between any two vertices there is at most one Directed edge.

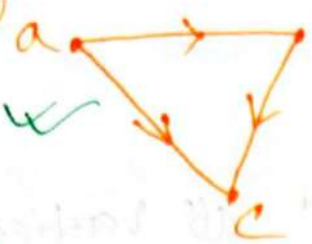
Ex:- ① ✓

② ✗

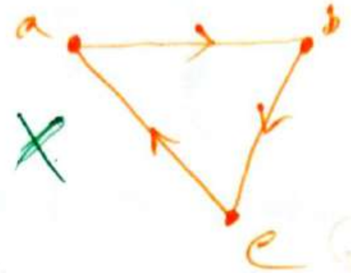
Transitivity:-

⇒ If (a,b) and (b,c) are edge, then so, is (a,c)

Ex:- (I)



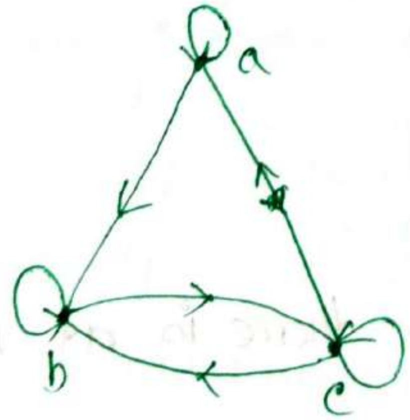
(II)



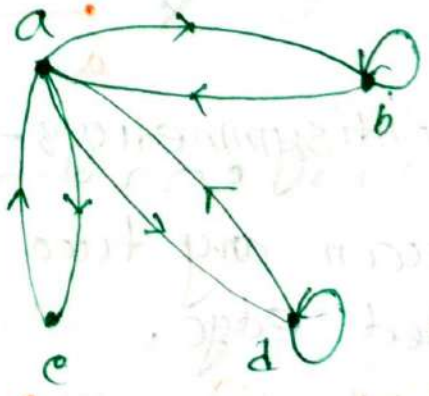
Problem: 10:-

⇒ Determine whether the relation for the directed graphs shown below are reflexive, symmetric, antisymmetric and/or transivite.

(1.)



(2.)



Ans: to the Q.no. 1

⇒ (1) Reflexive:- because there are loops at every vertex.

(2) Not Symmetric:- because there is an edge from a to b but not from b to a.

(3) Not antisymmetric:- because there is an edge from b to c and an edge from c to b.

(4) Not transitive:- because there is an edge from from a to b and edge from b to c but not edge from a to c.

Ans: to the Q.no. 2

(1) Not Reflexive:- because loops are not present every vertex.

(2) Symmetric:- Every edge between distint vertices is accompanied by edge in opposite direction.

(3) Not antisymmetric:- There is an edge from a to b and edge from b to a.

(4) Not transitive:- because (c, a) and (a, b) belongs to Relation but (c, b) does not belong to Relation.

Home Work:-

⇒ Represent each of this relations on $\{1,2,3,4\}$ with a matrix.

(a) $\{(1,2),(1,3),(1,4),(2,3),(2,4),(3,4)\}$

(b) $\{(1,1),(1,4),(2,2),(3,3),(4,1)\}$

(c) $\{(1,2),(1,3),(1,4),(2,1),(2,3),(2,4),(3,1),(3,2),(3,4),(4,1),(4,2),(4,3)\}$

(d) $\{(2,4),(3,1),(3,2),(3,4)\}$

(e) $\{(1,2),(1,3),(2,2),(2,1),(3,2),(3,4),(4,1)\}$

Ans: to: the: Q: no:-a

is given:-

$R = \{(1,2),(1,3),(1,4),(2,3),(2,4),(3,4)\}$

$A = \{1,2,3,4\}$

The matrix
$$M_R = \begin{array}{c|cccc|} & 1 & 2 & 3 & 4 \\ 1 & 0 & 1 & 1 & 1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & 0 & 0 & 1 \\ 4 & 0 & 0 & 0 & 0 \end{array}$$

Ans to! the Q!no!b

In given:-

$R = \{(1,1), (1,4), (2,2), (3,3); (4,1)\}$

$A = \{1, 2, 3, 4\}$

The matrix $M_R = \begin{array}{c|cccc|} & 1 & 2 & 3 & 4 \\ 1 & 1 & 0 & 0 & 1 \\ 2 & 0 & 1 & 0 & 1 \\ 3 & 0 & 0 & 1 & 0 \\ 4 & 1 & 0 & 0 & 0 \end{array}$

Ans to! the: Q:no:e

In given:-

$R = \{(1,2), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,1), (4,2), (4,3)\}$

$A = \{1, 2, 3, 4\}$

The matrix $M_R = \begin{array}{c|cccc|} & 1 & 2 & 3 & 4 \\ 1 & 0 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 1 & 0 \end{array}$

Ans to the Q no d

In given:—

$R = \{(2,4), (3,1), (3,2), (3,4)\}$

$A = \{1, 2, 3, 4\}$

$$\text{Matrix } M_R = \begin{array}{c|cccc|} & 1 & 2 & 3 & 4 \\ 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 0 & 0 & 0 & 0 \end{array}$$

Ans to the Q no e:—

In given:—

$R = \{(1,2), (1,3), (2,2), (2,1), (3,2), (3,4), (4,1)\}$

$A = \{1, 2, 3, 4\}$

$$\text{The Matrix } M_R = \begin{array}{c|cccc|} & 1 & 2 & 3 & 4 \\ 1 & 0 & 1 & 1 & 0 \\ 2 & 1 & 1 & 0 & 0 \\ 3 & 0 & 1 & 0 & 1 \\ 4 & 1 & 0 & 0 & 0 \end{array}$$

The pigeonhole principle:-

⇒ If k is a positive integer and $k+1$ or more object are placed into k boxes. Then there is at least one box containing two or more of the object.

① Corollary 1:-

⇒ A function from a set with $k+1$ or more elements is not one to one.

#problem: 1:-

⇒ How many people born is at least same birthday among any group of 367 people.

⇒ We know:-

1 year = 366 days

That means:- there are only 366 possible birthday.

So, Among any group of 367 people there must be at least 2/two with the same birthday.

#problem;3:—

⇒ How many student must be in a class to guarrantee the at least two students receive the Same score on the final exam, if the exam is graded on a scale from 0 to 100.

⇒ Is given:—

The exam is graded on a scale from 0 to 100 points.

So, There are 101 possible scores.

Now, The minimum number of students required $= 101+1$
$= 102$

That means:— If there are 102 or more students, we canbe sure that at least two of them will receive the same score.

Generalized pigeonhole principle:—

3) If n pigeonholes are occupied by $kn+1$ or more pigeons, where k is a positive integer, then at least one pigeonhole is occupied by $k+1$ or more pigeons.

1. If N objects are placed into K boxes, then there is at least one box containing $\lceil N/K \rceil$ objects

problem:-58—

3) Among the Number of 100 people. How many people are born in Same month.

is given:—

Total Number of people $N = 100$

Total Number of Month $K = 12$

∴ We know:—

among 100 people there are at least $\lceil \frac{N}{K} \rceil = \lceil \frac{100}{12} \rceil = 9$ who are born in the same month.

problem 6:—

⇒ What is the minimum number of student required in a DM class to be sure that at least six will receive the same grade if there is a 5 (five) possible grate.

⇒ is given:—

$$K = 5$$

We know:—

The minimum number of student required $N = nK + 1$

$$= 5 \cdot 5 + 1$$
$$= \underline{26}$$

because that at least six student receive the same grade is the smallest integer N such that

$$\left\lceil \frac{N}{5} \right\rceil = 6$$

$$\therefore N_{smallest} = \left\lceil \frac{26}{5} \right\rceil = 6$$

Problem: 31 :—

⇒ There are 38 diffrent time periods during which classes at a university canbe scheduled. If there are 677 different classes, how many diffrent rooms will be needed?

⇒ in given:—

$K = 38$

$N = 677$

So Diffrent rooms will be needed $= \left\lceil \frac{N}{K} \right\rceil = \left\lceil \frac{677}{38} \right\rceil$

$= \lceil 17.81 \rceil$

$= 18$ rooms

That means:—

⇒ 18 diffrent rooms will be needed for 38 diffrent times periods if there are 677 diffrent classes.

Lecture : 11

Matrices :-

⇒ A matrices is a rectangular array of numbers. A matrices with m rows and n colums is called an m×n matrix.

1. The plural of matrix is matrices.
2. A matrix with the same number of rows as colums is called Square matrix.
3. Two matrix are equal if they have the same Number of rows and the Same Number colums and the corresponding entries in every position are equal.

Example :-

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 0 & 4 \end{bmatrix}$$

The matrix is a 3×2 matrix.

Notation:

Addition of two matrices:

⇒ Let $A=[a_{ij}]$ and $B=[b_{ij}]$ be $m \times n$ matrices. The sum of A and B denoted by $A+B$ is the $m\times n$ matrix that has $a_{ij}+b_{ij}$ as its (i,j)th element.

Example:

$$A=\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$$

$$B=\begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$$

$$\therefore A+B=\begin{bmatrix} a_1+b_1 & a_2+b_2 \\ a_3+b_3 & a_4+b_4 \end{bmatrix}$$

Problem-1: $A=\begin{bmatrix} 1 & -2 & 3 \\ 4 & 5 & -8 \\ -10 & 6 & 5 \end{bmatrix}$, $B=\begin{bmatrix} -6 & 6 & 4 \\ -12 & -8 & 8 \\ 9 & 5 & -1 \end{bmatrix}$ add the two matrix.

$$\Rightarrow A+B=\begin{bmatrix} 1-6 & -2+6 & 3+4 \\ 4-12 & 5-8 & -8+8 \\ -10+9 & 6+5 & 5-1 \end{bmatrix}$$

$$=\begin{bmatrix} -5 & +4 & 7 \\ -8 & -3 & 0 \\ -1 & 11 & 4 \end{bmatrix}$$

Subtraction of 2(two) Matrices

⇒ The subtraction of two matrices of the same size is obtained by subtracting element in the corresponding positions.

(1.) Only matrices with the same dimensions can be added and subtracted.

(2.) Matrices of diffrent sizes cannot be added and subtracted.

Ex: $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}, B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$

$$\therefore A - B = \begin{bmatrix} a_1 - b_1 & a_2 - b_2 \\ a_3 - b_3 & a_4 - b_4 \end{bmatrix}$$

Problem: 2:- $A = \begin{bmatrix} 1 & -2 \\ -10 & 10 \end{bmatrix}$; $B = \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix}$ Subtracted the two matrices.

$$\Rightarrow \quad A - B = \begin{bmatrix} 1-10 & -2-20 \\ -10-30 & 10-40 \end{bmatrix}$$

$$= \begin{bmatrix} -9 & -22 \\ -40 & -30 \end{bmatrix}$$

Product of two matrices:-

⇒ Let A be $m \times k$ matrix and B be $k \times n$ matrix. The product of A and B denoted by AB is the $m \times n$ matrix with its (i, j)th element equal to the sum of the products of the corresponding entries from the ith row of A and the jth column of B.

1. If $AB = [c_{ij}]$ then $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}$.

2. A product of two matrices is defined only when the number of columns in the first matrix equals the number of rows of the 2nd matrix.

Ex- $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \\ a_5 & a_6 \end{bmatrix}$, ; $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$

$$\therefore AB = \begin{bmatrix} a_1b_1 + a_2b_3 & a_1b_2 + a_2b_4 \\ a_3b_1 + a_4b_3 & a_3b_2 + a_4b_4 \\ a_5b_1 + a_6b_3 & a_5b_2 + a_6b_4 \end{bmatrix}$$

Problem: 5:

$A = \begin{bmatrix} 2 & 3 \\ 5 & 6 \\ 7 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$ Find the product of two matrix.

$$\Rightarrow AB = \begin{bmatrix} 2\times4 + 3\times2 & 2\times3 + 3\times1 \\ 5\times4 + 6\times2 & 5\times3 + 6\times1 \\ 7\times4 + 1\times2 & 7\times3 + 1\times1 \end{bmatrix}$$

$$= \begin{bmatrix} 14 & 9 \\ 32 & 21 \\ 30 & 22 \end{bmatrix}$$

Transpose of a Matrix:-

⇒ Let $A=[a_{ij}]$ be an $m \times n$ matrix. The transpose of A denote by A^t is the $n \times m$ matrix obtained by interchanging the rows and colums of A.

(1) If $A^t=[b_{ij}]$, then $b_{ij}=a_{ji}$ for $i=1,2,\ldots,n$ and $j=1,2,\ldots,m$.

Eg:- The transpose of the matrix

$$A=\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix} = A^t$$

Problem: 7:- $A=\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ Find the transpose of matrix A.

$$\Rightarrow \quad A^t=\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Zero-One Matrix.

⇒ A matrix with entries that are either 0 or 1 is called zero-one matrix.

1. Represented by zero-one matrices are based on Boolean arithmetic.

2. $b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise.} \end{cases}$

3. $b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise.} \end{cases}$

Join and Meet of Zero-One Matrices:

Join:

⇒ The join of A and B is the zero-one matrix with (i,j)th entry $a_{ij} \vee b_{ij}$. The join of A and B is denoted by $A \vee B$.

Meet:

⇒ The meet of A and B is the zero-one matrix with (i,j)th entry $a_{ij} \wedge b_{ij}$. The meet of A and B is denoted by $A \wedge B$.

1. These operations are only possible when A and B have same size. Just as in the case of matrix addition.

problems 9:— Find the join and meet of the zero-one matrices.

$$A=\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix},\quad B=\begin{bmatrix}0 & 1 & 0\\ 1 & 1 & 0\end{bmatrix}$$

$\Rightarrow$

$$A\vee B=\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 0\end{bmatrix}$$

$$A\wedge B=\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$$

problems 10:— Find the join and meet of the zero-one matrices.

$$A=\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix},\quad B=\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix}$$

$\Rightarrow$ The join of A and B is:—

$$A\vee B=\begin{bmatrix}1\vee 1 & 0\vee 0 & 1\vee 1\\ 0\vee 0 & 1\vee 1 & 0\vee 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix}$$

The meet of A and B is:—

$$A\wedge B=\begin{bmatrix}1\wedge 1 & 0\wedge 0 & 1\wedge 1\\ 0\wedge 0 & 1\wedge 1 & 0\wedge 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}$$

Boolean Product of Zero-one Matrices:—

⇒ Let $A=[a_{ij}]$ be an $m\times k$ zero-one matrix and $B=[b_{ij}]$ be an $k\times n$ zero-one matrix. The Boolen product of A and B, denoted by $A\odot B$ is the $m\times n$ zero-one matrix with (i,j)th entry c_{ij} where $c_{ij}=(a_{i1}\wedge b_{1j})\vee(a_{i2}\wedge b_{2j})\vee\ldots\vee(a_{ik}\wedge b_{kj})$

Problems:—

⇒ Find the Boolean product of A and B where.

$$A=\begin{bmatrix}1 & 0\\ 0 & 1\\ 1 & 0\end{bmatrix},\quad B=\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 1\end{bmatrix}$$

⇒ The Boolean product $A\odot B$ is:—

$$A\odot B=\begin{bmatrix}(1\wedge 1)\vee(0\wedge 0) & (1\wedge 1)\vee(0\wedge 1) & (0\wedge 1)\vee(1\wedge 0)\\ (0\wedge 1)\vee(1\wedge 0) & (0\wedge 1)\vee(1\wedge 1) & (0\wedge 0)\vee(1\wedge 1)\\ (1\wedge 1)\vee(0\wedge 0) & (1\wedge 1)\vee(0\wedge 1) & (1\wedge 0)\vee(1\wedge 0)\end{bmatrix}$$

$$=\begin{bmatrix}1\vee 0 & 1\vee 0 & 0\vee 0\\ 0\vee 0 & 0\vee 1 & 0\vee 1\\ 1\vee 0 & 1\vee 0 & 0\vee 0\end{bmatrix}$$

$$=\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 1 & 0\end{bmatrix}$$

#Mathematical Induction:—

⇒ To prove that $p(n)$ is true for all positive integers. n, where $p(n)$ is a propositional function, we complete two steps:-

① BASIS STEP:—

⇒ We verify that $P(1)$ is true.

② InDuctive STEP:—

⇒ We show that the conditional statement $P(K) \to P(k+1)$ is true for all positive integers K

③ Inductive hypothesis:—

⇒ $P(K)$ is true.

④ $[P(1) \land \forall K (P(K) \to P(k+1))] \to \forall n\, P(n)$

☒ Way to proof:—

① Basis:—

⇒ Took the 1st value of the statement from the left side and put the value of n in right hand side.

② Induction:— put $n=k$ and assume its true
put $n=k+1$ and show its true.

Problem 1

⇒ Show that if n is a positive integer, then $1+2+\cdots+n = n(n+1)/2$

⇒ Basis:

$n=1$

$1 = \frac{1(1+1)}{2}$

$1 = \frac{1+1}{2}$

$1 = 1$

L.H.S = R.H.S

So, $P(1)$ is true.

$n \geq 1$

⇒ Induction:

→ put $n=k$ and Assume its true.

$\therefore 1+2+\cdots+k = k(k+1)/2$ ——— (i)

→ Now put $n=k+1$ to shown that $P(k+1)$ is true.

$1+2+\cdots+k+1 = (k+1)(k+1+1)/2$

$\boxed{1+2+\cdots+k} + k+1 = (k+1)(k+2)/2$ ——— (ii)

Substitute the value of eq (i) in eq (ii) ⇒

$$\frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$$

$$\Rightarrow \frac{k(k+1)+2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$$

P.T.O

$\therefore k(k+1) + 2(k+1) = (k+1)(k+2)$

$(k+2)(k+1) = (k+1)(k+2)$

$\therefore (k+1)(k+2) = (k+1)(k+2)$

L.H.S = R.H.S.

Since, LHS = RHS so this statement is true.

So, we have prove that the above statement is true by mathematical Induction by using Basis and induction methods

Problem: 3

⇒ Use mathematical induction to show that:

$1+2+2^2+\cdots+2^n = 2^{n+1}-1$ for all nonnegative integers n.

⇒ Basis step:–

$1 = 2^{0+1}-1$ $\quad\{ n \geq 0$

$1 = 2^1 - 1$

$\therefore 1 = 1$

$\therefore$ L.H.S = R.H.S

So. $P(1)$ is true.

⇒ Induction:—

Let, $n = k$, & Assume that its true.

So, $1+2+2^2+\cdots+2^k = 2^{k+1}-1$ —— (I)

Now: put $n=k+1$ and show that its true:—

So, $1+2+2^2+\cdots+2^{k+1} = 2^{k+1+1}-1$

$\boxed{1+2+2^2+\cdots+2^k}+2^{k+1} = 2^{k+2}-1$ —— (II)

Now:- Substitute the value of eq (I) in eq (II) ⇒

$2^{k+1}-1+2^{k+1} = 2^{k+2}-1$

$2^{k+1}(1+1)-1 = 2^{k+2}-1$

$2^{k+1}\times 2 = 2^{k+2}-1+1$

$2^{k+1+1} = 2^{k+2}$

$\therefore 2^{k+2} = 2^{k+2}$

L.H.S = R.H.S

So, this statement in true.
prove that by Mathematical inductions steps.

Problems 3:-

⇒ Use mathematical induction to show that
$1^2+2^2+\cdots+n^2=n(n+1)(2n+1)/6$ for all positive integers n.

⇒ Basis:-

$$1^2=1(1+1)(2\times1+1)/6 \quad \Big| \quad n \geq 1$$

$$1=\frac{2\times3}{6}$$

$$1=1$$

L.H.S = R.H.S

So, $P(1)$ is true.

⇒ induction:-

⇒ Let, $n=k$ & Assume that the statement is true:

So, $1^2+2^2+\cdots+k^2=k(k+1)(2k+1)/6$ ——— (1)

Now, put $n=k+1$ and show that it's true:-

$$1^2+2^2+\cdots+(k+1)^2=(k+1)(k+1+1)(2(k+1)+1)/6$$

$$1^2+2^2+\cdots+k^2+(k+1)^2=(k+1)(k+2)(2k+3)/6 \quad \text{——— (11)}$$

Substitute the value of eq(1) in eq(11) ⇒

$$k(k+1)(2k+1)/6+(k+1)^2=(k+1)(k+2)(2k+3)/6$$

$$\frac{k(k+1)(2k+1)+6(k+1)^2}{6} = \frac{(k+1)(k+2)(2k+3)}{6}$$

$$k(k+1)(2k+1)+6(k+1)^2 = (k+1)(k+2)(2k+3)$$

$$(k+1)\{k(2k+1)+6(k+1)\} = (k+1)(k+2)(2k+3)$$

$$(k+1)\{2k^2+k+6k+6\} = (k+1)(k+2)(2k+3)$$

$$(k+1)\{2k^2+7k+6\} = (k+1)(k+2)(2k+3)$$

$$(k+1)\{2k^2+4k+3k+6\} = (k+1)(k+2)(2k+3)$$

$$(k+1)\{2k(k+2)+3(k+2)\} = (k+1)(k+2)(2k+3)$$

$$\therefore (k+1)(2k+3)(k+2) = (k+1)(2k+3)(k+2)$$

$$L.H.S = R.H.S$$

So Therefore the inductive step is true.

proved by mathematical induction steps.

Problem: 58

⇒ Use mathematical induction to prove that the inequality $n < 2^n$ for all positive integers n

⇒ Basis:

$1 < 2^1$

$1 < 2$

$\therefore P(1)$ is true.

⇒ Induction:

Let, $n = k$ and assume that the statement is true.

So, $k < 2^k$ ——— (I)

Now! put $n = k+1$ and show that its true:

$(k+1) < 2^{(k+1)}$

——— (II)

Now: We need to show that if $k < 2^k$, then $k+1 < 2^{k+1}$

Then: Multiply 2 to both sides of eq (I) ⇒

$2k < 2^k \cdot 2$

$2k < 2^{k+1}$

$k + k < 2^{k+1}$ ——— (III)

'k' in positive integer,

for $n=1$, it in true.

So, $k>1$

$k+k>1+k \longrightarrow$ [Add k in both side] — (IV)

Compaire the equation (III) and (IV) :—

$1+k<k+k<2^{k+1}$

$\therefore (1+k)<2^{k+1}$

$\therefore (k+1)<2^{(k+1)}$

So, it in true for $n=k+1$ when $n=k$ in true

Hence prove by Mathematical Induction $2^n>n$ in true.

problem-58

⇒ Prove that $1^2+3^2+5^2+\dots+(2n+1)^2=(n+1)(2n+1)(2n+3)/3$ whenever n is a nonnegative integer:

⇒ Basis:—

$$1^2=(0+1)(2\times0+1)(2\times0+3)/3$$

$$1=(1+0)(1+0)(3+0)/3$$

$$\therefore 1=\frac{3}{3}.$$

$$1=1$$

L.H.S = R.H.S

$\therefore$ $P(1)$ is true.

⇒ Induction:—

Let, $n=k$ and assume that is true.

$$1^2+3^2+5^2+\dots+(2k+1)^2=(k+1)(2k+1)(2k+3)/3 \quad \text{—— (1)}$$

Now, put $n=k+1$ and prove that its true:—

$$1^2+3^2+5^2+\dots+\{(k+1)\times2+1\}^2=(k+1+1)(2(k+1)+1)\{2(k+1)+3\}/3$$

$$1^2+3^2+5^2+\dots+\{(2k+3)\}^2=(k+2)(2k+3)(2k+5)/3$$

$$1^2+3^2+5^2+\dots+(2k+1)^2+(2k+3)^2=(k+2)(2k+3)(2k+5)/3$$

Substitute the Value of eq(1) in eq (11) $\Rightarrow$

$$(k+1)(2k+1)(2k+3)/3 + (2k+3)^2 = (k+2)(2k+3)(2k+5)/3$$

$$\Rightarrow \frac{(k+1)(2k+1)(2k+3)+3(2k+3)^2}{3} = (k+2)(2k+3)(2k+5)/3$$

$$\Rightarrow (2k+3)\{(k+1)(2k+1)+3(2k+3)\} = (k+2)(2k+3)(2k+5)$$

$$\Rightarrow (2k+3)\{2k^2+k+2k+1+6k+9\} = (k+2)(2k+3)(2k+5)$$

$$\Rightarrow (2k+3)\{2k^2+9k+10\} = (k+2)(2k+3)(2k+5)$$

$$\Rightarrow (2k+3)\{2k^2+4k+5k+10\} = (k+2)(2k+3)(2k+5)$$

$$\Rightarrow (2k+3)\{2k(k+2)+5(k+2)\} = (k+2)(2k+3)(2k+5)$$

$$\Rightarrow (2k+3)(2k+5)(k+2) = (2k+3)(2k+5)(k+2)$$

$\therefore$ L.H.S = R.H.S

So, its true for $n=k+1$ when $n=k$ in true.

Hence proved by MI the Statement in true.

Lecture 8.13:-

#Directed Graph:-

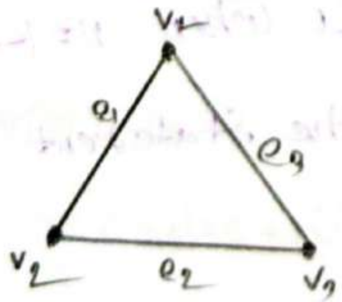
⇒ A directed graph (digraph) consists of a nonempty set of vertices V and a set of directed edges E.

(1) The directed edge associated with the ordered pair(u,v) is said to start at u and end at v.

#Types of Graphs:-

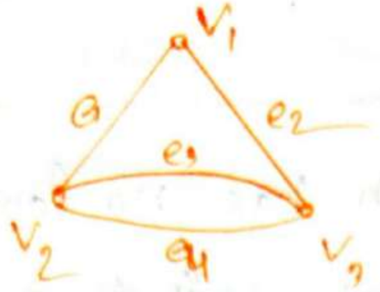
Simple graph:-

⇒ A graph does not contain any self loop and multiedge.



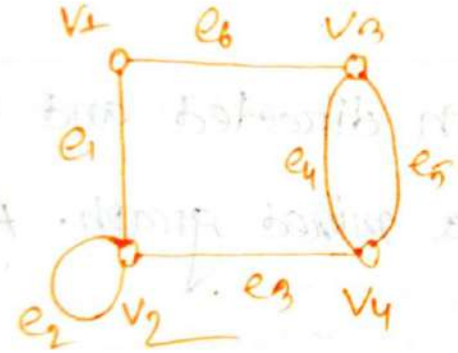
Multigraph:-

⇒ A graph does not contain any self loop but contain multiedge is called multigraph.



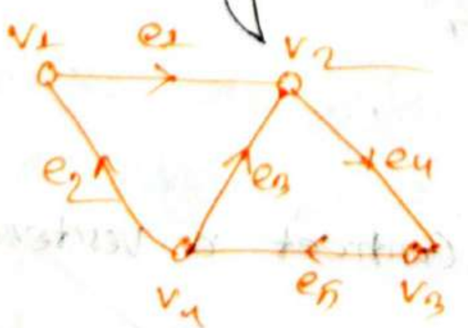
☒ Pseudo Graphs —

⇒ A graph contain both self loop and multiedge is called pseudo graph.



☒ Directed graph:—

⇒ A graph consist the direction of edges then this is called directed graph.

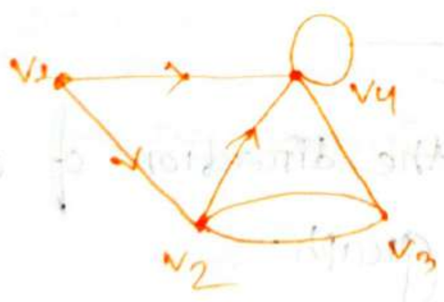


Simple Directed graph:—

⇒ When a directed graph has no loop and has no multiple directed edges, it in called simple directed graph.

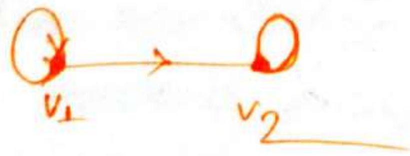
Mixed graph:—

⇒ A graph with both directed and underricted edges in called a mixed graph. A mixed graph may contain loops.



Loop:—

⇒ An edge that contreet a vertex to itself in called a loop.

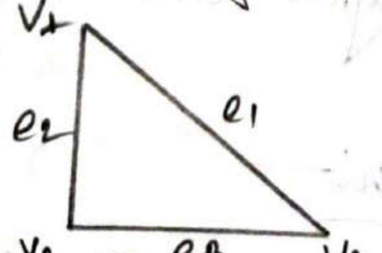


Type	Edges	Multiple Edges Allowed	Loops Allowed
Simple graph	undirected	NO	NO
multigraph	undirected	yes	NO
Pseudograph	undirected	yes	yes
Simple directed graph	Directed	No	No
Directed Multi-graph	Directed	yes	yes
mixed graph	unDirected Directed	yes	yes.

Incidence and Adjacency:—

⇒ Let e_k be an edge joining two vertices V_i and V_j. Then e_k is said to be incident of V_i and V_j

⇒ Two vertices are said to be adjacent if there exist an edge joining this vertices.



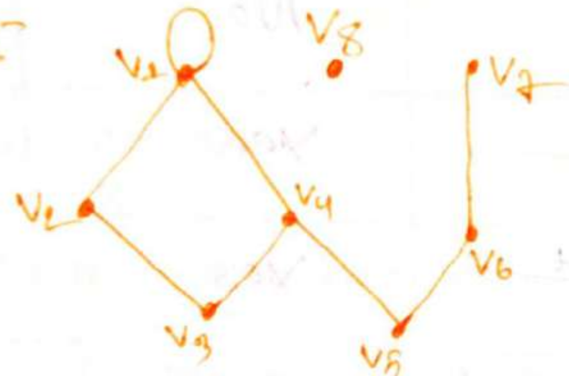
1. Here e_1 is incident of V_1 and V_2. V_1 and V_2 are adjacent but V_3 and V_4 are not adjacent.

Degree of Vertex

⇒ The Degree of vertex v in a graph G written as d(v) is equal to the number of edges are incident on v. Self loop counted twice.

denote by deg(v)

Ex:-



here, $d(v_1) = 4$, $d(v_2) = 2$,
$d(v_3) = 2$, $d(v_4) = 3$,
$d(v_5) = 2$, $d(v_6) = 2$,
$d(v_7) = 1$
$d(v_8) = 0$

☒ Isolated vertex:—

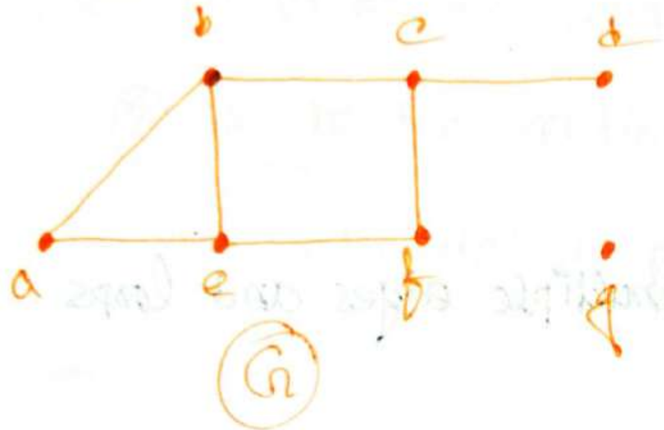
⇒ A vertex of degree zero is called Isolated.

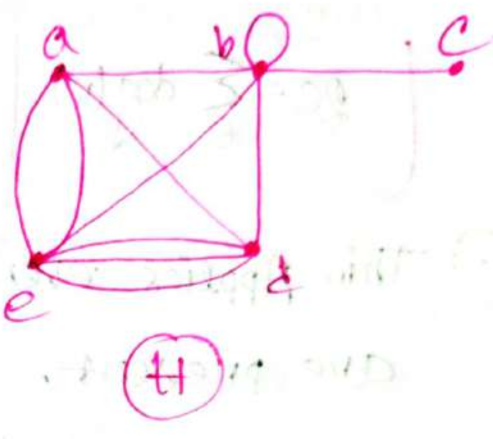
☒ Pendant Vertex:—

⇒ A vertex having degree 1 is called Pendant vertex.

Problem: 1:

⇒ What are the degrees of the vertices in the graph G and H?





⇒ Degrees of graph G is:

deg(a) = 2, deg(b) = 3, deg(c) = 3

deg(d) = 1, deg(e) = 3, deg(f) = 2

deg(g) = 0

⇒ Degrees of graph H is:

deg(a) = 4

deg(b) = 6

deg(c) = 1

deg(e) = 6

deg(d) = 5

The Handshaking Theorem:-

⇒ Let, $G=(V,E)$ be an undirected graph with e edges then,

$$\boxed{2e = \sum_{E} deg(v)}$$

(1.) This Applies even if multiple edges and loops are present.

problem 2:-

⇒ How many edeges are there in a graph with 10 vertices each of degree six?

⇒ In given:-

$V = 10$; dig of 1 vertex $= 6$

$\therefore d(V) = 10 \cdot 6 = 60$

we know:-

$$2e = \sum deg(v)$$

$$\therefore e = \frac{60}{2}$$

$$= 30$$

$\therefore$ 30 edeges are in the graph.

Ⓐ An Undirected graph has a even number of Vertices of odd degree:—

⇒ Let, (1.) V_1 be the vertices of even degree and

(2) V_2 be the vertices of odd degree.

in an undirected graph $G=(V,E)$ with m edges,

Then,

$$2m = \sum \deg(v) = \sum \deg(v_1) + \sum \deg(v_2)$$

problem: 03:—

⇒ If a graph has 5 Vertices, can each Vertex has degree 3.

⇒ In given:—

$V = 5$

each Vertex degree $= 3$

$\therefore$ total number of degree $= 3 \cdot 5$

$= 15$ is a odd

This is not possible by the Hand-Shaking theorem.

Inital Vertex and Terminal Vertex:-

⇒ When (u,v) is an edge of the graph G with directed edges,

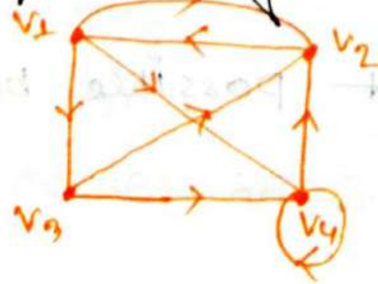
1. u is said to be Adjacent to v
2. v is said to be Adjacent from u
3. The vertex u is called initial vertex of (u,v)
4. The vertex v is called terminal/End vertex of (u,v)
5. The initial vertex and the terminal vertex of a loop are the same.

In-Degree and Out-degree of a Vertex:-

In-Degree:-

⇒ Number of incoming vertex, denote by $deg^{-}(v)$

Ex:-



$deg^{-}(v_1) = 1$

$deg^{-}(v_2) = 3$

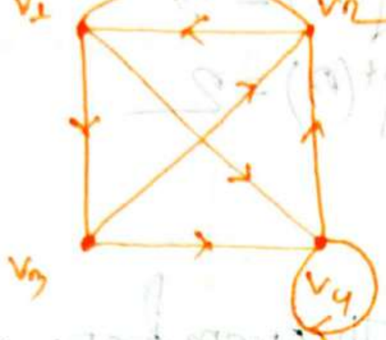
$deg^{-}(v_3) = 1$

$deg^{-}(v_4) = 3$

Out-Degree:—

⇒ Number of out-going vertex. denote by $deg^+(v)$

Ex:—



$deg^+(v_1) = 2$

$deg^+(v_2) = 1$

$deg^+(v_3) = 2$

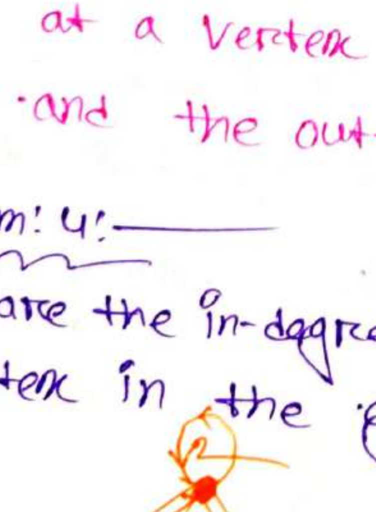
$deg^+(v_4) = 2$

Note:—

⇒ A loop at a vertex contributes 1 to both the indegree and the out-degree of this vertex.

problem: 4:—

⇒ What are the in-degrees and out-degress of all the vertex in the graph below?



⇒ Indegree:—

$deg^-(1) = 0$

$deg^-(2) = 3$

$deg^-(3) = 4$

⇒ Out-Degrees:—

$deg^+(1) = 2$

$deg^+(2) = 3$

$deg^+(3) = 2$

Handshaking Theorem for Directed graph:—

⇒ Let $G = (V, E)$ be a graph with directed edges, Then

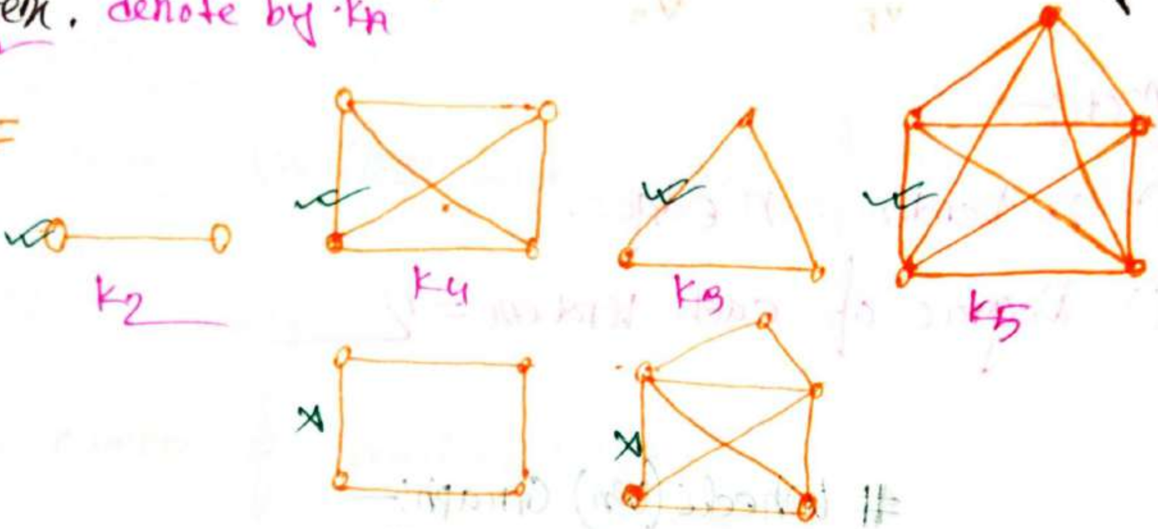
$$\sum_{\in} deg^-(v) = \sum_{\in} deg^+(v) = |E|$$

Complete Graph: (K_n)

⇒ A simple connected graph is said to be complete graph if each vertex is connected to every other vertex. denote by K_n

Ex:-

(1).

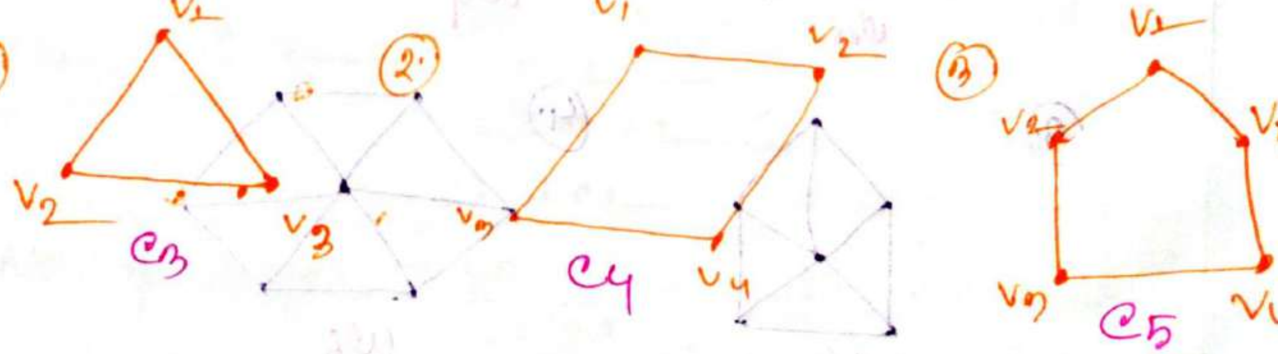


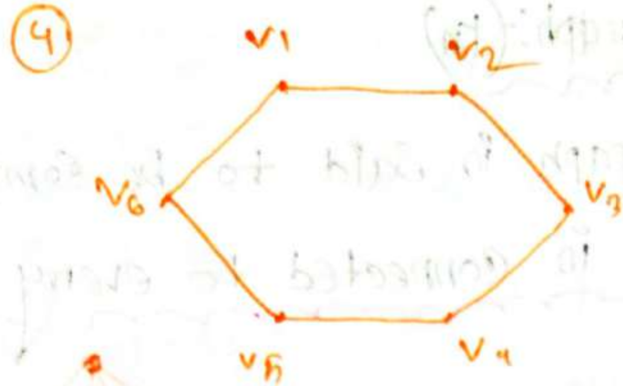
Cycles (C_n): Graph:—

⇒ The cycle C_n, $n \geq 3$ consists of n vertices $v_1, v_2, \dots, v_n$ and edge $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$ and $\{v_n, v_1\}$

⇒ A graph containing atleast one cycle in it is known as cycle graph. denote by C_n

Ex:-



(4) 

Note:-

(i) n Vertex = n edges.

(ii) Degree of each vertex = 2

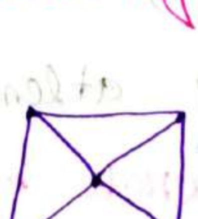
Wheels (W_n) Graph:-

⇒ The wheel W_n is obtained when an additional Vertex to the cycle C_n for $n \geq 3$ and connect this new vertex to each of the n vertices in C_n by new edges. denote by W_n.

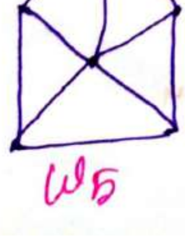
Ex:-

(1) 

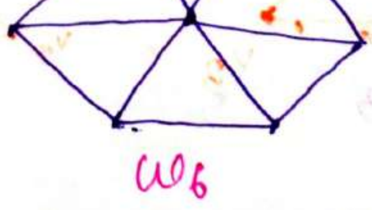
W_3

(2) 

W_4

(3) 

W_5

(4) 

W_6

Note:— $W_n \Rightarrow$

(i) n+1 vertices.

(ii) 2n edeges.

Problem:—

⇒ How many vertices and edges of W_4 wheels:?

⇒ in given:—

$n=4$

∴ number of vertices $= n+1$
$= 4+1$
$= 5$

∴ number of edeges $= 2n$
$= 2*4$
$= 8$

Problem:—

⇒ How many edeges an vertices of W_{100} wheels:?

⇒ In give:—

$n=100$

∴ Number of vertices $= n+1$
$= 100+1$
$= 101$

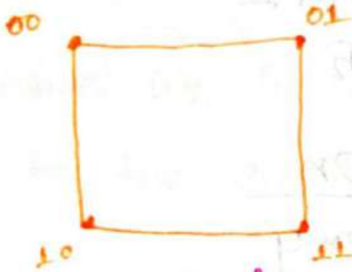
∴ No. of edges $= 2n$
$= 2*100$
$= 200$

n-Cubes (Q_n) graph:—

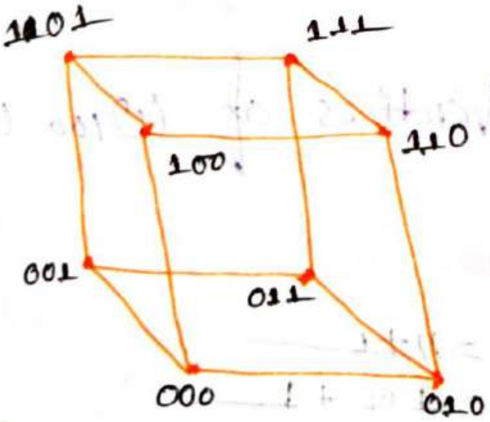
⇒ The N-Cubes graph in the graph that has Vertices representing the 2^n bit string of lenght n. denote by Q_n

Ex:- (1) $Q_1 \Rightarrow n=1$ ∴ no. of Vertices $= 2^n = 2^1 = 2$

(2) $Q_2 \Rightarrow n=2$ ∴ no. of vertices $= 2^n = 2^2 = 4$



(3) $Q_3 \Rightarrow n=3$ no. of vertices $= 2^n = 2^3 = 8$



Note:—

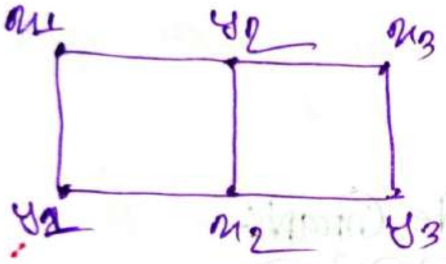
1. Two Vertices adjacent iff the bit strings that they represent differ in exactly one bit position
2. Edege $= n \cdot 2^{n-1}$
3. degree $= n$ regular graph.

Bipartite Graph:-

⇒ A graph G is said to be a bipartite graph if its vertex set V can be partitioned into two sets, say V_1 and V_2, such that no two vertices in the same partition can be adjacent.

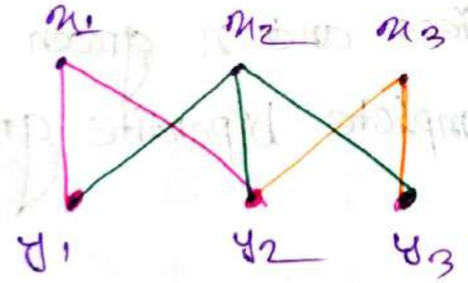
Here the (V_1, V_2) is called the bipartition of G

Ex:-



Can its called Bipartite graph or not?

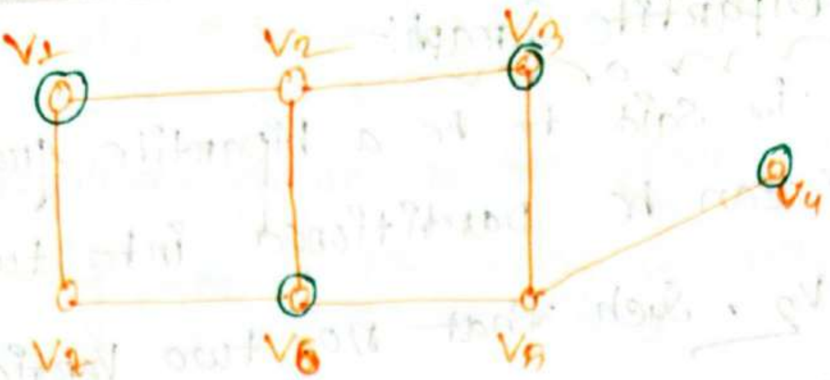
⇒ Re Draw the graph:-



$\therefore V_1 = \{x_1, x_2, x_3\}$

$V_2 = \{y_1, y_2, y_3\}$

Ex:-



Here, $V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$

$X = \{v_1, v_6, v_3, v_4\}$

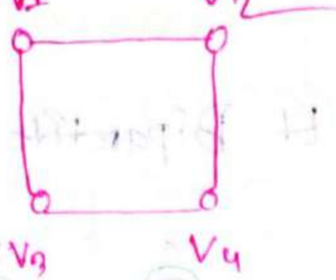
$Y = \{v_2, v_7, v_5\}$

Complete Bipartite Graph:-

⇒ When all possible edeges exist in a sample bipartite graph with m red Vertices and n green Vertices, The graph is called complete bipartite graph. denote by $K_{m,n}$.

⇒ If every Vertices in X is ~~disjoint in~~ every Vertices in Y, the it's called a complete bipartite graph if x and y condition in m and n Vertices then this grap is denote by $K_{m,n}$.

Ex:- 1.



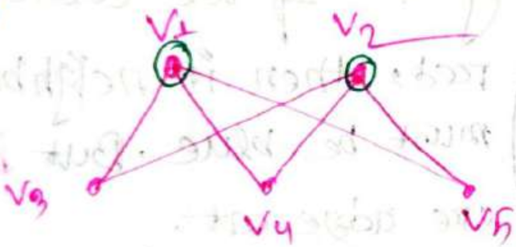
Here, $V = \{v_1, v_2, v_3, v_4\}$

$X = \{v_1, v_2\}$

$Y = \{v_2, v_3\}$

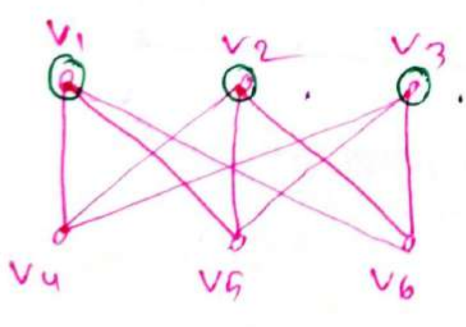
This graph is $K_{2,2}$

2.



$K_{2,3}$

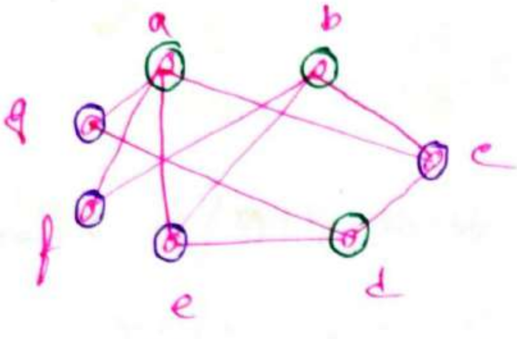
3.

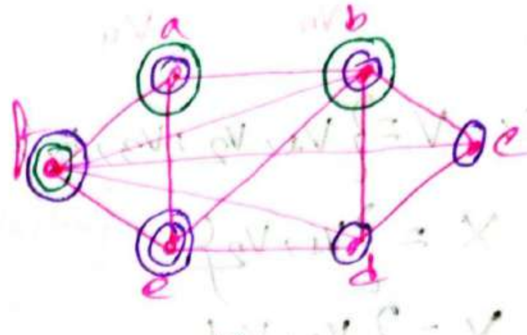


$K_{3,3}$

Probeem:-

⇒ Are the graph G and H Bipartite or not?.





⇒

$x = \{a, b, d\}$

$y = \{g, f, e, c\}$

So, the graph G is bipartite.

⇒ ∴ H is not a bipartite graph. If we color a red, then it neighbors b, f must be blue. But b and f are adjecent.

Lecture 26:-

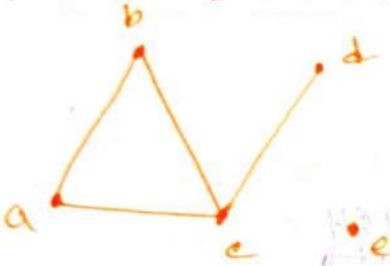
Representing Graphs:-

Adjacency lists:-

⇒ A table with 1 row per vertex, listing its adjacent vertices.

Undirected Adjacency lists:-

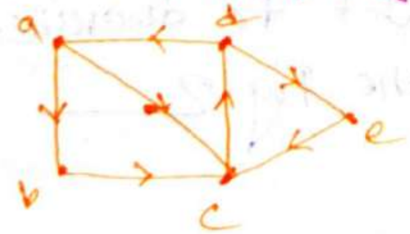
Ex:-



Vertex	Adjacency List
a	b, c
b	a, c
c	b, a, d
d	c
e	∅

Directed Adjacency Lists:-

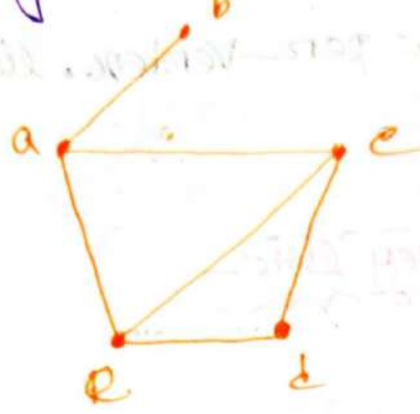
Ex:-



Vertex	Adjacency List
a	b, c
b	c
c	d
d	a, e
e	c

Problems:

⇒ Use Adjacency list to describe the simple graph given in Figure 1.

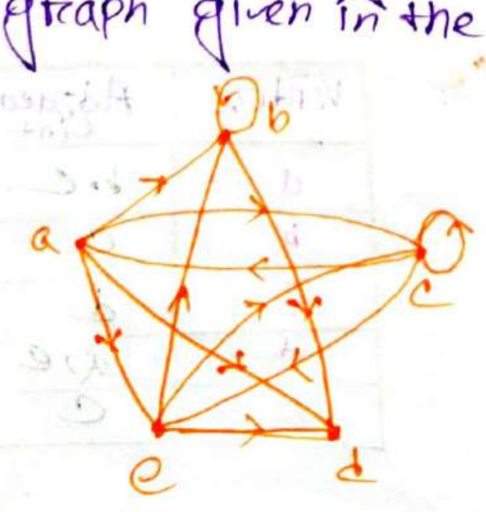


⇒

Vertex	Adjacency List
a	b, c, e
b	a,
c	a, d, e
d	c, e
e	a, c, d

Problem: 2:

⇒ Use Directed Adjacency list to describe the Directed graph given in the Fig 2



Vertex	Adjacency list
a	b, c, d, e
b	b, d
c	a, c, e
d	ϕ
e	b, c, d

Representing Graphs Using Matrices:-

⇒ There are two types of matrices.

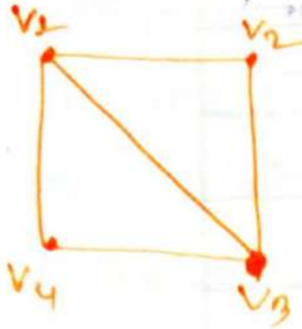
1. Adjacency matrix.
2. Incidence matrix,

Adjacency Matrix:- (Square)

⇒ A matrix representing a graph using the adjacency of vertices.

1. All undirected graps, including multigraps and Psedougraps, have Symmetric adjacency matrix.

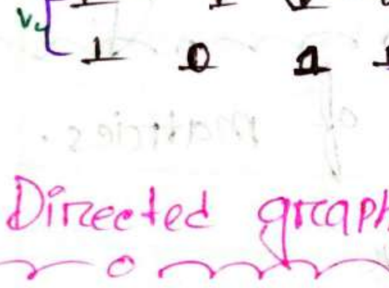
(1.) For Undirected graph:—



$\therefore$ Adjacency matrix:—

$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

(2.) For Directed graph:—

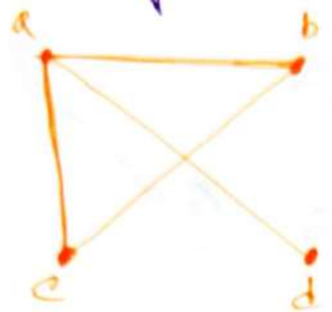


$\therefore$ Adjacency matrix:—

$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Problem: 3:-

⇒ Use an adjacency matrix to represent the graph Shown in the Fig.

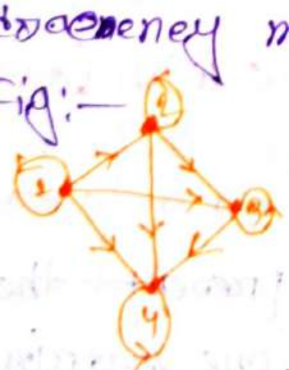


⇒ The adjacency matrix is:-

$$A = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Problem 4:-

⇒ Use an adjacency matrix to represent the graph shown in Fig:-



⇒ The Adjacency matrix is:-

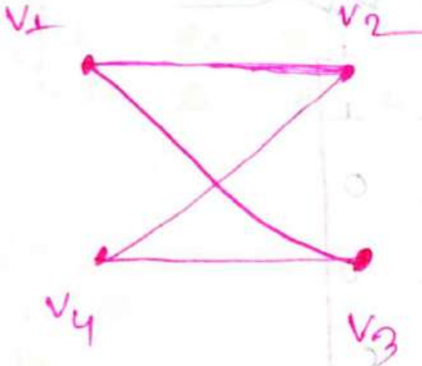
$$A = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Problem: 4!—

⇒ Draw a Graph with a the adjacency Matrix below.

$$\begin{array}{c|cccc} & v_1 & v_2 & v_3 & v_4 \\ \hline v_1 & 0 & 1 & 1 & 0 \\ v_2 & 1 & 0 & 0 & 1 \\ v_3 & 1 & 0 & 0 & 1 \\ v_4 & 0 & 1 & 1 & 0 \end{array}$$

⇒ The graph is:—

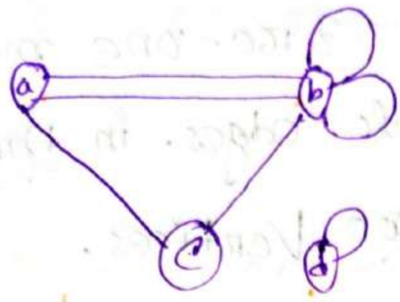


(1.1) For Undirected graphs with loops and Multiple edges:—

⇒ When multiple edges are present the Adjacent matrix is no longer zero-one matrix.

#problem: 1 :-

⇒ whats the adjacency matrix of the following graph?



⇒ The adjacency matrix is:-

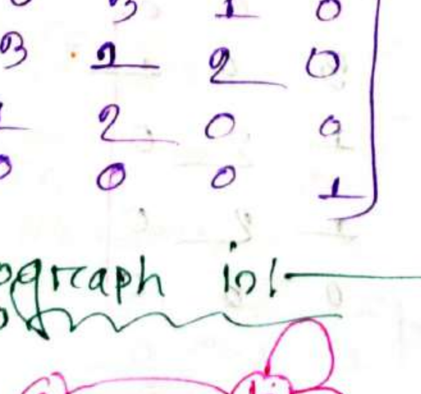
$$A = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & & & & \\ b & & & & \\ c & & & & \\ d & & & & \end{array} \left[\begin{array}{cccc} 0 & 2 & 1 & 0 \\ 2 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Problem: 2 :-

⇒ Use an adjacency matrix to represent the pseudograph.

$$A = \begin{bmatrix} 0 & 3 & 1 & 0 \\ 3 & 2 & 2 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

⇒ The pseudograph is:-

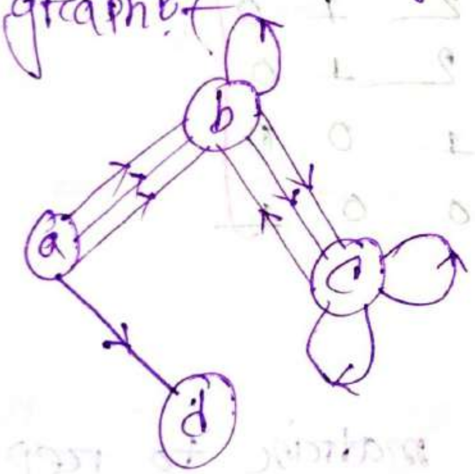


Q.1 For Directed Graph with loops and Multiple edges:-

⇒ The matrices are not zero-one matrices when there are multiple edges in the same direction connecting two vertices.

problem: 10-

⇒ What are the adjacency matrix of the following graph?



⇒ The Adjacency matrix:-

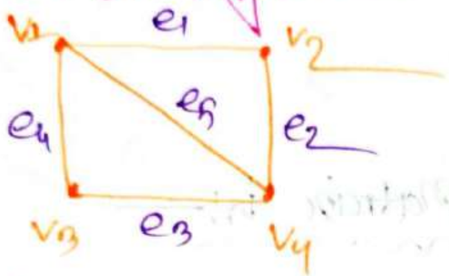
$$A = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 3 & 0 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Incidence Matrices:- (V×E) matrix

⇒ A matrix representing a graph using the incidence of edges and vertices.

For Undirected graph:-

Ex:-

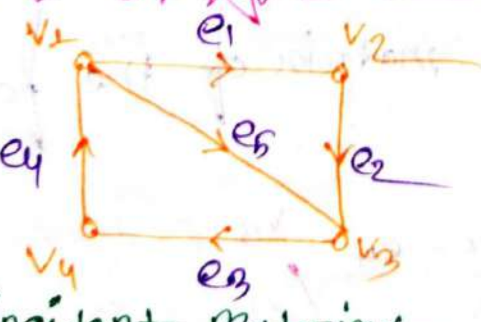


⇒ The incidence matrix:-

$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

For Directed graph:-

Ex:-



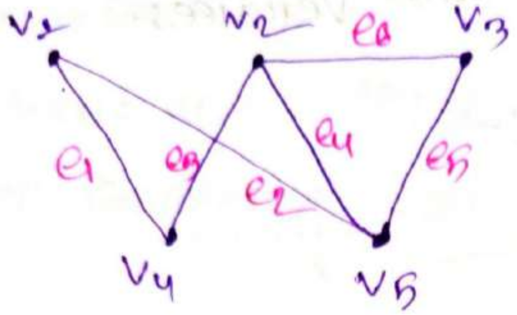
For Incoming = −1
Outgoing = 1
null = 0

The incident matrix,-

$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix} \end{array}$$

Problem: 6

⇒ Whats the incident matrix of the following graph.

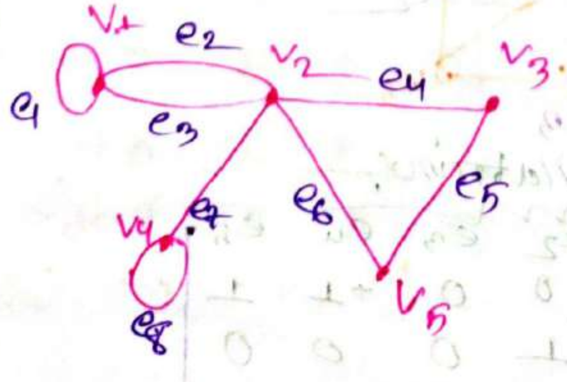


⇒ The Incident Matrix is:—

$$A = \begin{array}{c|cccccc} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \hline v_1 & 1 & 1 & 0 & 0 & 0 & 0 \\ v_2 & 0 & 0 & 1 & 1 & 0 & 1 \\ v_3 & 0 & 0 & 0 & 0 & 1 & 1 \\ v_4 & 1 & 0 & 1 & 0 & 0 & 0 \\ v_5 & 0 & 1 & 0 & 1 & 1 & 0 \end{array}$$

Problem: 7

⇒ Whats the incident matrix of the following graph:—



⇒ The incident matrix is:—

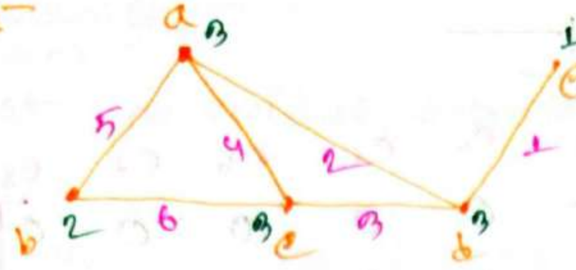
$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{cccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Isomorphism of Graphs:—

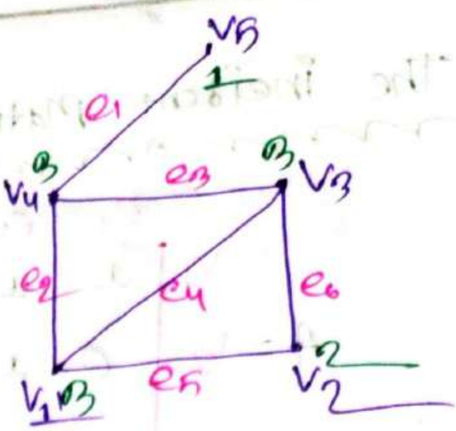
⇒ Two graphs G_1 and G_2 are said to be isomorphic to each other if there is a one-to-one correspondence (bijection) between their vertices and between there edges such that the incidence releationship is preserved.

1. Two graph G_1 and G_2 are isomorphic if
   (i) Number of Vertices are same.
   (ii) Number of Edges same.
   (iii) An equal number of vertices with given degree.
   (iv) Vertex Correspondance and edge Correspondance valid.

Ex:-



G_1



G_2

Sol:-

(1) Number of Vertices $G_1 = G_2 = 5$

(2) Number of edeges $G_1 = G_2 = 6$

(3) $DegSeq(G_1) = 3, 3, 3, 2, 1$

$DegSeq(G_2) = 3, 3, 3, 2, 1$

(4.1) Vertex Correspondance:-

$f(a) = V_1$

$f(b) = V_2$

$f(c) = V_3$

$f(d) = V_4$

$f(e) = V_5$

(4.2) Edeges Correspondance:-

$f(1) = e_1$

$f(2) = e_3$

$f(3) = e_2$

$f(4) = e_4$

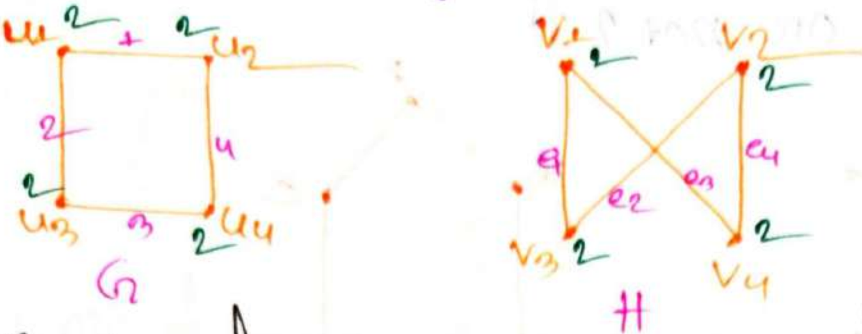
$f(5) = e_5$

$f(6) = e_6$

$\therefore$ Adjacency Also preserved. So G_1 and G_2 are said to be Isomorphic.

Problem: 8!

⇒ Show that the graphs G and H are Isomorphic.



⇒ ① Number of Vertices $G = H = 4$

② Number of edeges $G = H = 4$

③ $DegSeq(G) = 2, 2, 2, 2$

$DegSeq(H) = 2, 2, 2, 2$

(4.1) Vertex Corrlarespondance:-

$f(u_1) = v_1$

$f(u_2) = v_2$

$f(u_3) = v_3$

$f(u_4) = v_4$

(4.2) Edege Corrrespondance:-

$f(1) = e_1$

$f(2) = e_2$

$f(3) = e_3$

$f(4) = e_4$

∴ Adjacency also perserved so, the Graph G and H Isomorphic.

Problem: 0!—

⇒ Show that the following graphs G and H are Isomorphic or not?

⇒ (i) No. of Vertices G = H = 5

(ii) No. of Edeges G = 6 but

No. of " H = 5

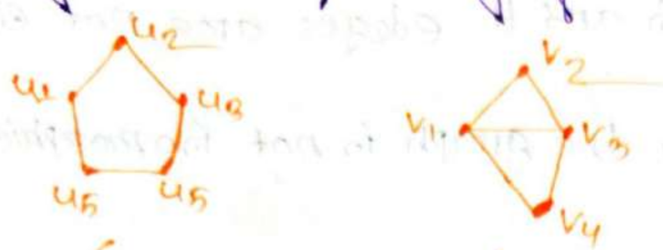
We know:—

For a isomorphic graph No. of edeges are equal but in the graphs G and H edeges are not equal.

Sol: The graph is not Isomorphic.

Problem: 1!—

⇒ Why are the following graphs not Isomorphic?



G H

⇒ (1) No. of vertices of Graph $G = 5$ but
No. of " of " $H = 4$

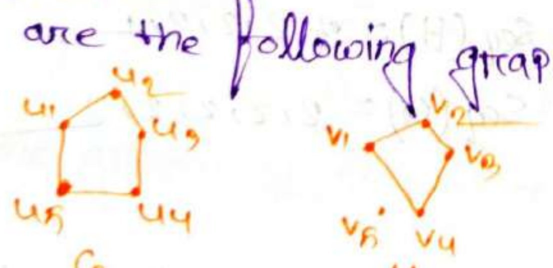
We know:—

⇒ For a Isomorphic graphs No. of vertices of both graphs must be same. but in the graphs G and H are not equal.

So, The graph is not Isomorphic.

Problem: 2!—

⇒ Why are the following graphs not isomorphic?



G H

⇒ (1.) No. of vertices of graphs $G = H = 5$

(2.) (i) No. of edges of graph $G = 5$

(ii) No. of " " " $H = 4$

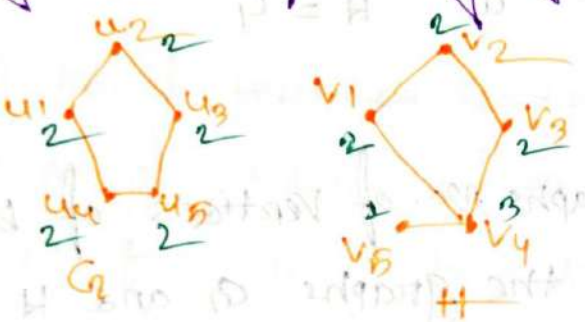
We know:—

⇒ For a Isomorphic graphs no. of edeges are equal. but in the Graphs G and H edeges are not equal.

So, That's the reason the graph in not isomorphic.

Problem: 3:—

⇒ Why are the following graphs not isomorphic?



⇒ (1) The no. of Vertices $G = H = 5$

(2) The no. of edeges $G = H = 5$

(3) The no. of Degree $Seq(H) = 3, 2, 2, 2, 1$

" " " $Seq(G) = 2, 2, 2, 2, 2$

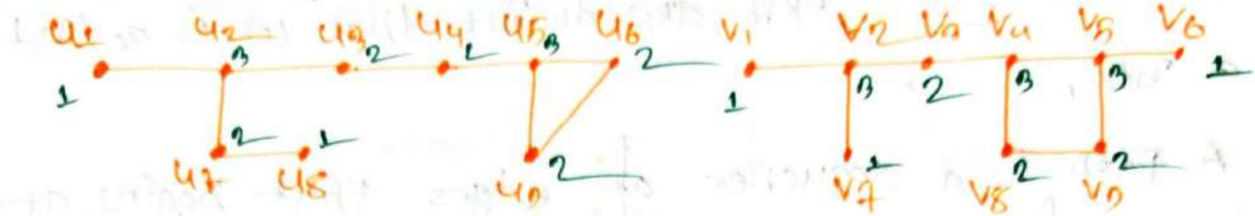
We know:—

⇒ For Isomorphic graph the equal number of Vertices with same degree but in the graph H has vertex of degree 1, in the graph G doesn't has no vertex of degree 1.

So, That's the reason the graphs are not Isomorphic.

Problem: 4:-

The following graphs are isomorphic or not?



G_1 G_2

⇒ Check the all condition of a Isomorphic graph:-

(1) No. of Verticies = $G_1 = G_2 = 9$

(2) No. of edeges = $G_1 = G_2 = 9$

(3) (i) Degree Seq $(G_1) = 3, 3, 2, 2, 2, 2, 2, 1, 1$

(ii) Degree Seq $(G_2) = 3, 3, 3, 2, 2, 2, 1, 1, 1$

We know:-

For a Isomorphic graph the equal number of vertex with same degree. In the graph(G_1) 2 Vertices has degree 3, but In the graph (G_2) 3 Vertices has degree 3.

So, The graphs are not Isomorphic.

Lecture: 17:-

Path:-

⇒ If the vertices $v_0, v_1, \dots, v_k$ of the walk $w = v_0 e_1 v_1 e_2 v_2 \dots e_k v_k$ are distint then w is called a path.

⇒ A path is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of the graph.

1. Edege no repeat
2. Vertex no repeat

Walk:-

⇒ A walk in a graph G is a finite sequnce

$$W = v_0 e_1 v_1 e_2 v_2 e_3 \dots v_{k-1} e_k v_k$$

whos terms are alternately vertices and edeges such that, for $1 \le i \le k$, the edge e_i has ends v_{i-1} and v_i

Circuit:-

⇒ If it begins and end at the same vertex then the path is a circuit.

1. A path/circuit is simple it does not contain the same edge more than one.

   1. Vertex can repeat but
   2. Edege no repeat

Paths in Directed Graph:—

⇒ Same as in undirected graphs, but the Pat must go in the direction of the arrows.

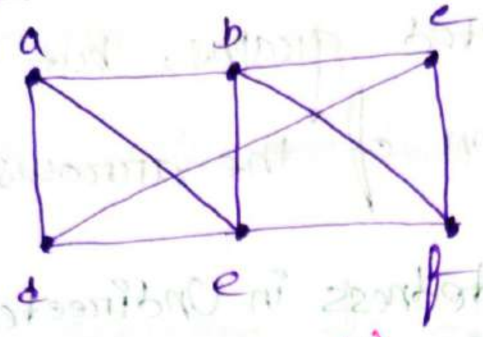
Connectedness in Undirected Graph:—

⇒ An Undirected graph is called connected if there is a path between every pair of distinct Vertices of the graph.

Length of a path:—

⇒ The Summation of all the edegs traveled is called length of Path.

Example:- (1)



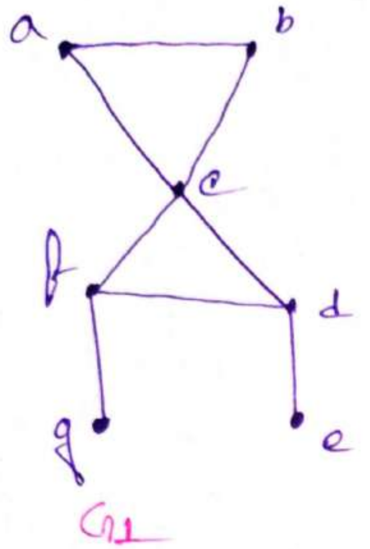
(1.) In the graph a, b, c, d, e is a simple path of length = 4 because {a, b}, {b, c}, {c, d}, {d, e} are all edge.

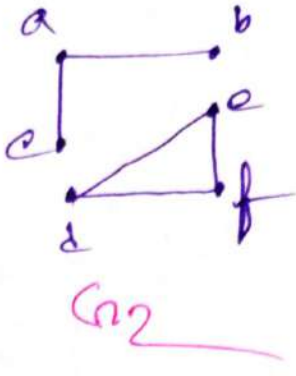
(2.) d, e, c, a is not a path, because {e, c} is not an edge.

(3.) b, c, f, e, b is a circuit of length 4. because {b, c}, {c, f}, {f, e}, {e, b} are edges and this Path begins and end at b.

(4.) The walk a, b, e, d, a, b which is of length 5, is not a simple because it contains the edge {a, b} twice.

Examples 6:-





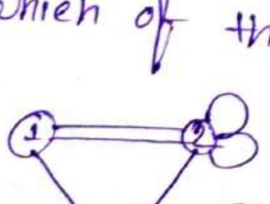
(1.) The graph G1 is connected. For every pair of distint vertices there is a path between them.

(2.) The graph G2 is not connected.
because there is a no path between the pair of vertices (c, d) and (b, e).

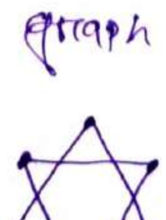
problem-1:

⇒ Which of the following graph are connected?

(1)



(2.)



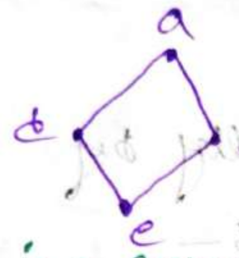
(3.)

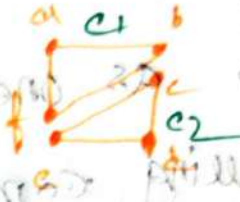
⇒ 1st and 2nd one is disconnected. But last one is connected.

Connected Component of Graph:-

⇒ A maximal connected subgraph of G is a component of G

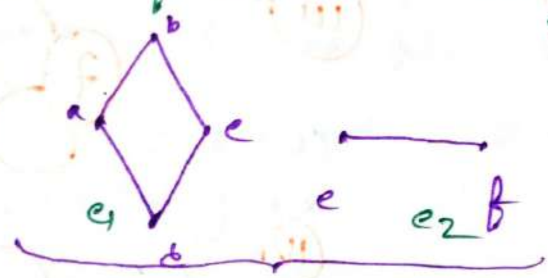
Ex:-





this is 1 connected component of Graph

this is 2 component where e_1 and e_2 are the subgraph

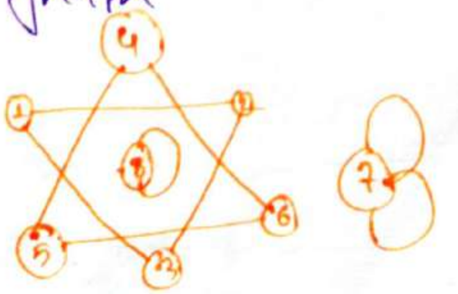


G_1

this graph have 2 connected component where e_1 and e_2 are the subgraph.

Problem: 1:-

⇒ What are the connected components of the following graph



⇒ The connected components are

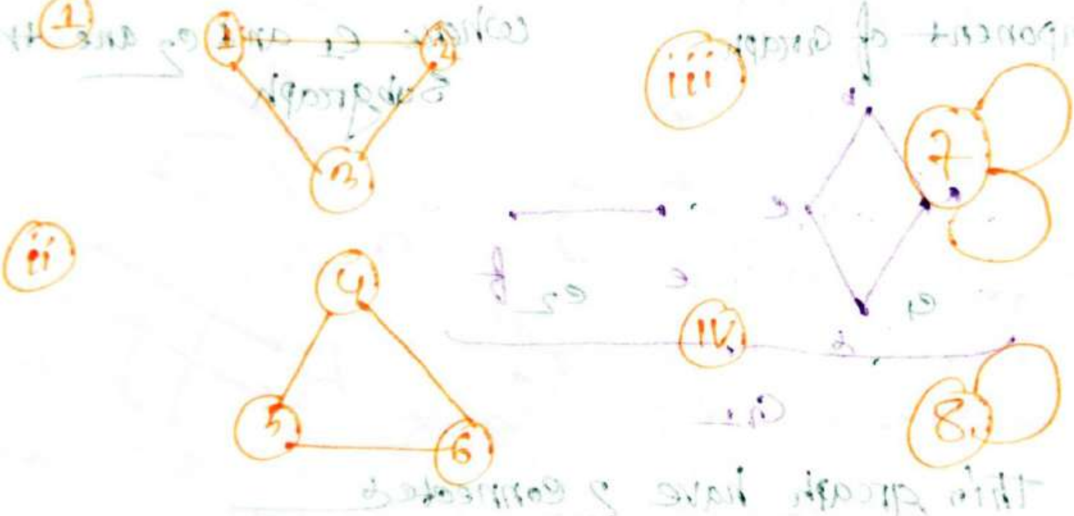
(i) {1, 2, 3}

(ii) {4, 5, 6}

(iii) {7}

(iv) {8} as one can see visually by

Pulling components apart:—



Connectedness in Directed graph:-

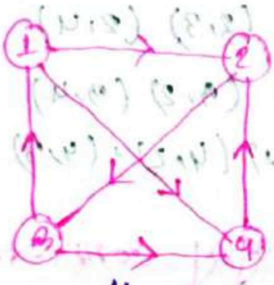
① Strongly connected.

② Weakly connected.

Strongly connected:-

⇒ A directed graph is called strongly connected graph if for any pair of vertices of the graph both the vertices of the pair are rechable from one another.

Ex:-



H

is the graph strongly connected or not?.

Pair of vertices	Forword path	Reverse pair	Backword Path:-
(1,2)	1-2	(2,1)	2-3-1
(1,3)	1-4-3	(3,1)	3-1
(1,4)	1-4	(4,1)	4-2-3-1
(2,3)	2-3	(3,2)	3-1-2
(2,4)	2-3-4	(4,2)	4-2
(3,4)	3-4	(3,4)	4-2-3.

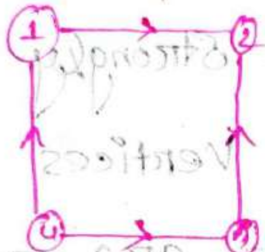
∴ H is strongly connected graph because there is a Path between any two vertices in the directed graph

Hence, H is weakly connected.

Wekly Connected or connected:-

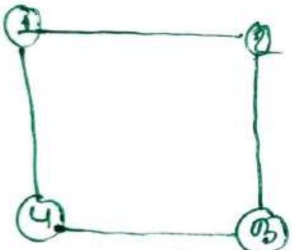
⇒ A directed graph is called Connected or weakly Connected if it is connected as an undirected graph in which each directed edge Connected to an undirected graph.

Ex:-



Thinking Connected graph or not?

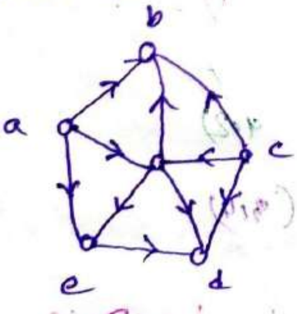
②



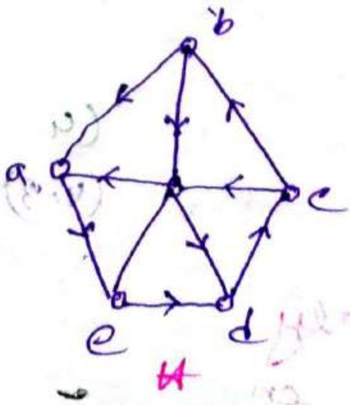
(1,2) (1,4) ; (1,3)
(2,1) (2,3) (2,4)
(3,1) (3,2) (3,4)
(4,1) (4,2) , (4,3)

⇒ The graph is not strongly connected. There is no directed path from a vertex to other vertex in the graph.

Problem-①



G



H

Q:- which of the following graph is strongly connected or not?

⇒ For graph: G

→ The graph G is not strongly connected because, There is no directed path form vertex b to other vertex in this graph.

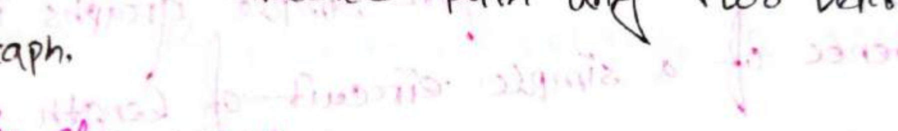
⇒ For graph H:

→ The graph H is strongly connected. because, there is a directed path any two vertices in this directed graph.

Hence, H is also weakly connected.

Problem:- 2:-

which of the following Graph is strongly connected and which one is connected.



⇒ For graph: G:-

→ This is strongly connected graph, because there is a directed path any two vertices in this directed graph.

⇒ For graph H:-

→ This is not strongly connected graph because there is a no directed graph from vertex a to other vertex in this graph.

Path of Isomorphic Graph:—

⇒ There are several ways that paths and circuits can help determine whether two graphs are isomorphic

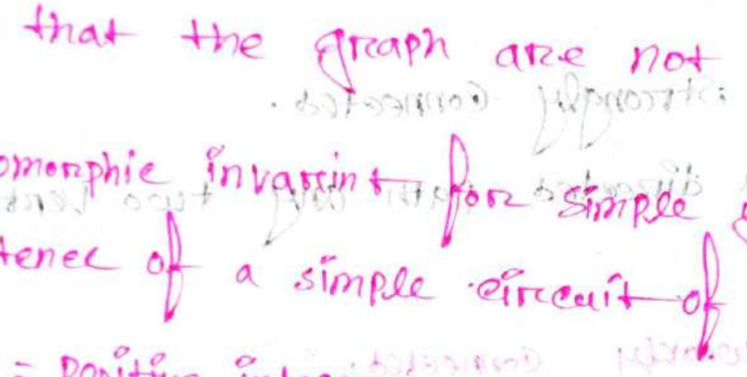
(1) The existence of a simple circuit of a particular length is a useful invariant that can be used to show that the graph are not Isomorphic.

(2) Isomorphic invariant for simple graphs is the existence of a simple circuit of length K.

K = positive integer ≥ 2.

Problem:— 1:—

⇒ Determine whether the graph G and H are Isomorphic ?



⇒ Check 1st 3 condition of Isomorphic graph if there is a any mistake then the graph is not Isomorphic.

(1.) No. of vertices $G = H = 5$.

(2) No. of edges $G = H = 5$

(3) Deg seq $(G) = 2, 2, 2, 2, 2$

Deg Seq $(H) = 2, 2, 2, 2, 2$

(4.) After that we will check the possible simple Circuit with length. If there in any possible simple circuit with length are mismatch between two graph the we say that the invariants are not same.

That means the graph in not Isomorphic.

⇒ Here, All the invariants are same.

(5.) After that now, we go for mapping:

$f(v_1) = u_1$

$f(v_2) = u_2$

$f(v_3) = u_3$

$f(v_4) = u_4$

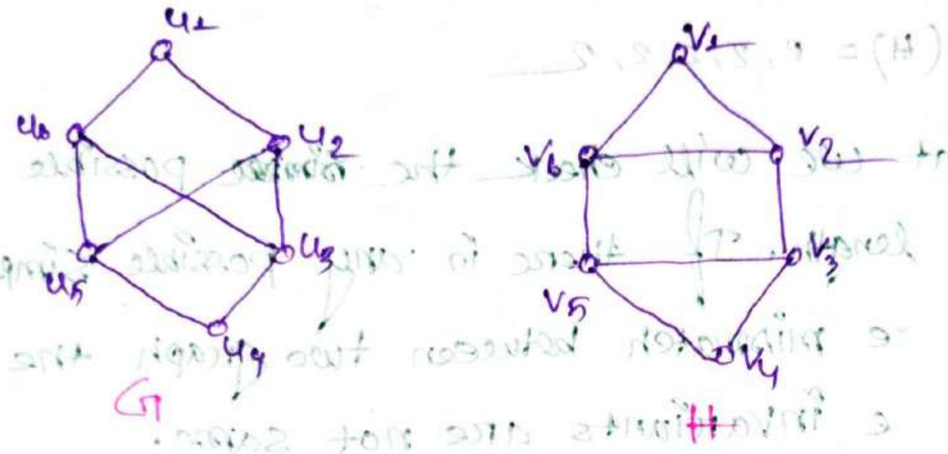
$f(v_5) = u_5$

So, the function in a bijective.

That means:- The graphs are Isomorphic.

Problem: 14:—

⇒ Determine whether the graphs G and H shown in the figure are Isomorphic or not.



⇒ Check all condition for Isomorphic:—

(1) No. of vertices G = H = 6

(2) No. of edges G = H = 8

(3) Deg Seq G = 3, 3, 3, 3, 2, 2
Deg Seq H = 3, 3, 3, 3, 2, 2

So the three invariants are same.

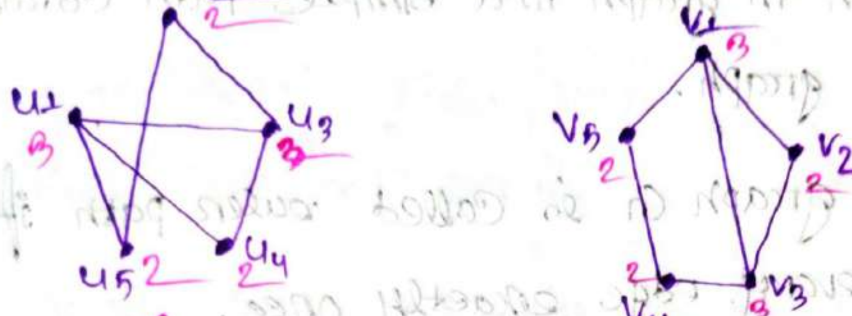
(4) Now, H has a Simple Circuit of length three there in v_1, v_2, v_6, v_1) but G has no Simple Circuit of length three.

Because the existence of a simple circuit of length three in an Isomorphic invariant,

So, G and H are not Isomorphic.

Problem: 18:

⇒ Determine whether the graps G and H in figure isomorphic or not.



G H

⇒ Check the all Condition of Isomorphic:

(1) No. of Verties $= G = H = 5$

(2) N.o of edeges $G = H = 6$

(3) Deg Seq $(G) = 3, 3, 2, 2, 2$

Deg Seq $(H) = 3, 3, 2, 2, 2$

So, The three invarcats are same.

Now,

(5) Both have three Simple circuit of length three and two Simple Circuit of length four. and one Simple circuit of length Five.

Because:- All these isomorphic invarciants agree, so G and H maybe Isomorphic.

maping:- $f(u_1) = v_1$; $f(u_4) = v_2$

$f(u_2) = v_4$; $f(u_5) = v_5$

$f(u_3) = v_3$; ∴ the function is bijective.

The graph is Isomorphic.

Lecture: 18:—

Eular Pats and Circuits:—

Euler path:—

⇒ An Euler path in graph is a simple path Cotaining every edge of graph.

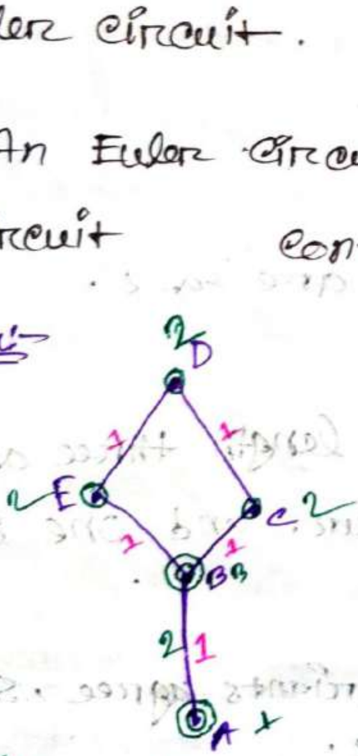
⇒ A path in a graph G is called euler path if it includes every edge exactly once.

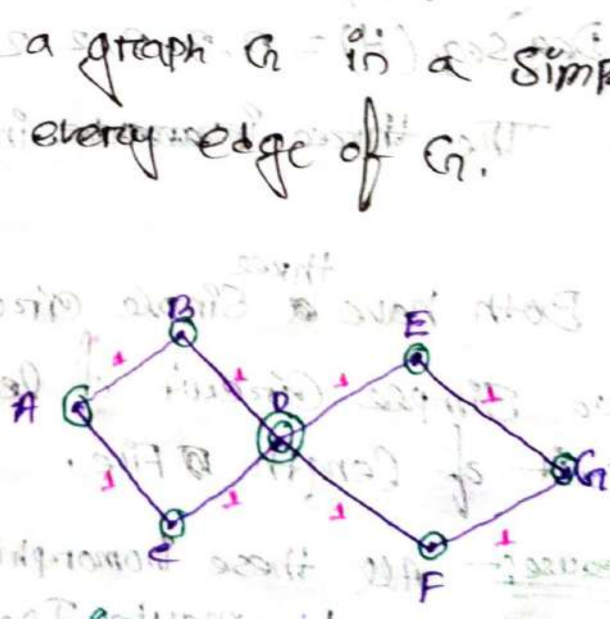
Euler Circuit:—

⇒ An euler path that is Circuit is called Euler circuit.

⇒ An Euler Circuit in a graph G is a simple Circuit containing every edge of G.

Ex:—





it's a euler path but not euler circuit.

it's a euler path also a euler circuit

Namely:— A, B, C, D, E, B

Namely:— A, B, D, F, G, E, D, C, A

Condition of Euler paths and circuits:-

(1) A graph contains Euler circuit if all of its vertices have even degree.

(2) A graph contains Euler path if exactly two of its vertices have odd degree.

(3) A graph does not contain any euler path or any Euler circuit if the graph have more than two vertices of odd degree.

(4) (i) If a graph contain Euler circuit, it also contain a euler path.

(ii) If a graph contain Euler path, maybe or maybe not it contain a euler circuit.

(5) An euler path is a path that uses every edge of a graph exactly once.

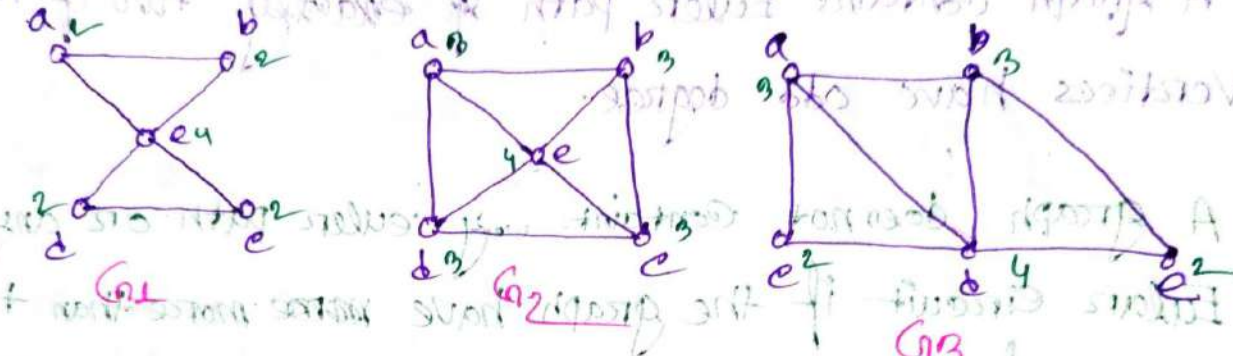
(6) An euler circuit is a circuit that uses every edge of a circuit exactly once.

(7) An euler path starting vertex and end vertex are different or equal.

(8) An euler circuit start and ends at the same vertex.

Problem 1:

⇒ Which of the undirected graphs in Figure below an euler circuit? of those that do not, which have Euler path?



⇒ For Graph G1:—

→ The graph G1 has an Euler circuit and also has an euler path namely a, b, c, d, e, a

⇒ For graph G2:—

→ The graph G2 Neither a has an Euler path or Euler circuit.

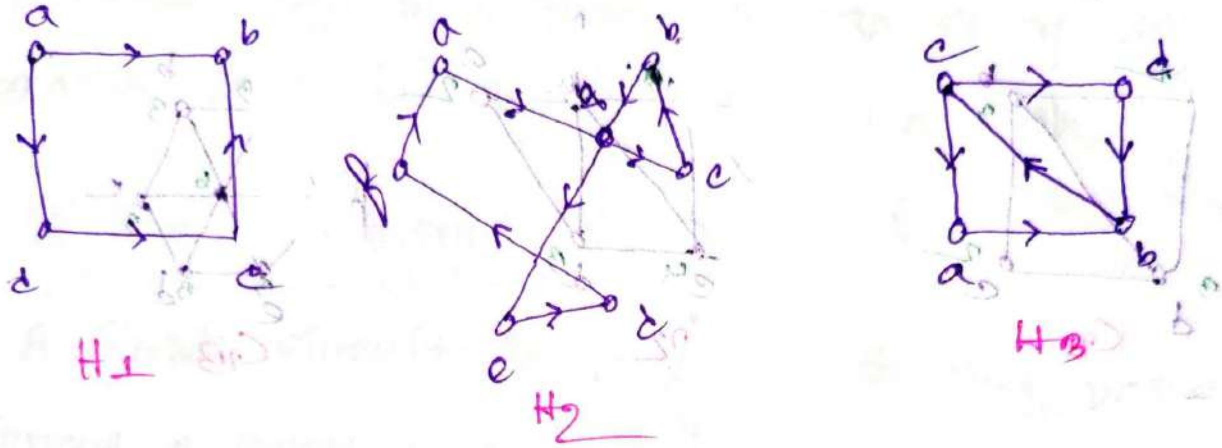
⇒ For graph G3:—

→ The graph G3 has an Euler path namely a, c, d, a, b, d, e, b, G3 does not have Euler circuit.

because the G3 graph contain exactly two vertices of odd degree.

Problem: 2 :-

⇒ which of the directed graphs in Figure have an Euler Circuit. of those that do not have Euler path?



⇒ For H1 graph:-

→ The graph H1 Neither has an Euler path or Euler Circuit.

⇒ For H2 graph:-

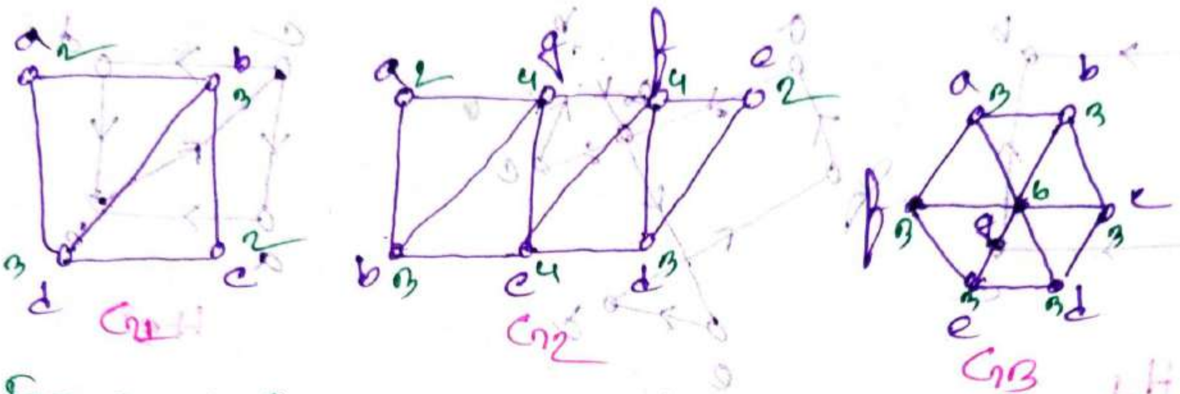
→ The graph H2 has an Euler Circuit also has an euler Path Namely, a,g,c,b,g,e,d,f,a

⇒ For H3 graph:-

→ The graph H3 has an Euler path Namely c,a,b,c,d,b but H3 does not have euler circuit

Problem: 4:-

⇒ Which graph shown in the figure have an Euler path:



⇒ For graph G_1:-

→ The graph G_1 has an Euler path namely
b, a, d, b, c, d
because G_1 contains exactly two vertices of odd degree namely b and d

⇒ For graph G_2:-

→ The graph G_2 has an Euler path namely
b, a, g, b, c, g, f, c, d, f, e, d,
because G_2 has exactly two vertices of odd degree namely b, d.

⇒ For graph G_3:-

⇒ The graph G_3 has no euler path. because it has six vertices of odd degree which is greater than 2

Hamilton Paths and Circuits:—

Hamilton path—

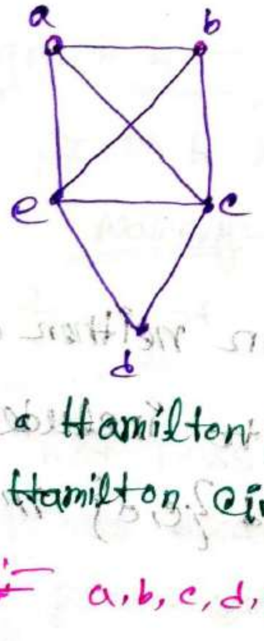
⇒ A simple path in a graph through every vertex exactly once is called a Hamilton path.

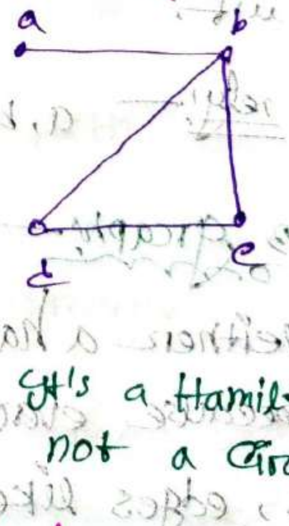
Hamilton Circuit:—

⇒ A simple circuit in a graph G that passes through every vertex exactly once is called Hamilton circuit.

Note:— A Hamilton path/circuit does not necessarily pass through all the edges of the graph.

Ex:—



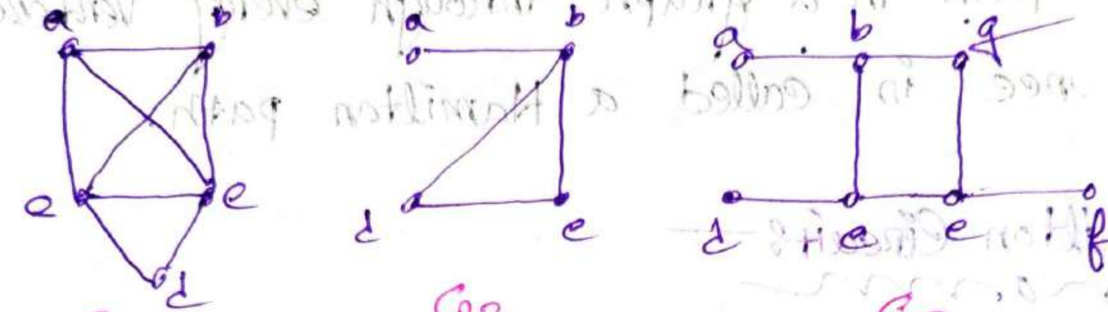


It's a Hamilton path & also a Hamilton circuit.
Namely:— a, b, c, d, e, a

It's a Hamilton path but not a circuit.
Namely:— a, b, c, d

Problem 5:—

⇒ Which of the simple graph in Figure have a Hamilton circuit or, if not a Hamilton Path?



G_1 G_2 G_3

⇒ For G_1 graph:—

⇒ It's a hamilton path and also a hamilton circuit.

Namely · a, e, d, c, b, a

⇒ For G_2 graph:—

⇒ It's has a hamilton path but Nethier has a hamilton circuit.

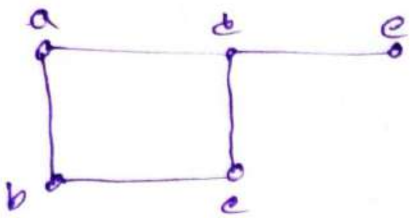
Namely:— a, b, e, d

⇒ Fo G_3 graph:—

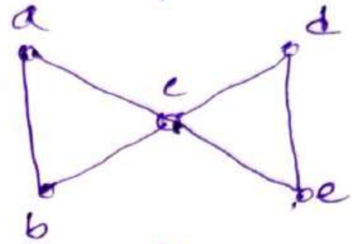
⇒ G_3 Neither a hamilton circuit or neither a hamilton Path, because every paths needs to include all the Vertices, edges like {a, b}, {e, f} and {e, d} more than once.

Problem: 6!—

⇒ Show that neither graph displayed in the Figure has a Hamilton circuit. Is there any Hamilton path in the graph.



G



H

⇒ For graph G:—

⇒ The graph G has an hamilton path

Namely:— a, b, c, d, e

But does not have a hamilton circuit.

because:—

G has a Vertex of degree one,

namely:— e.

⇒ For graph H:—

⇒ The graph H has an hamilton path

Namely:— a, b, c, d, e

But does not contain hamilt circuit

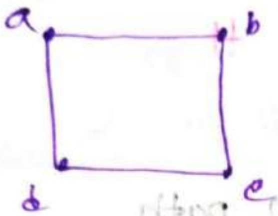
Because:—

It is not possible to start a point, visit all the other points exactly once. And then come back to the Starting point.

DIRAC'S THEOREM:—

If G is a simple graph with n vertices with $n>=3$ such that the degree of every vertices in G is at least $n/2$, then G has a Hamilton circuit.

Ex:—



$\therefore n=4\ (n\geq 3)$

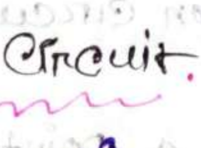
$\frac{n}{2}=\frac{4}{2}=2$

deg(a) = 2,
deg(b) = 2
deg(c) = 2
deg(d) = 2.

ORE'S THEOREM:—

⇒ If G is a simple graph with n vertices with $n>=3$ such that, $deg(u)+deg(v)>=n$ for every pair of non-adjacent vertices u and v in G, then G has a Hamilton circuit.

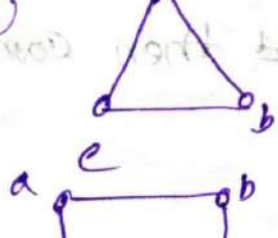
Ex:— (i)



$n=3\ (n\geq 3)$

✗

(ii)



$n=4\ (4\geq 3)$ true

a pair of non-adjacent vertices (a,c)

$\therefore deg(a)+deg(c)=2+2\geq n$

$=4\geq 4$ true

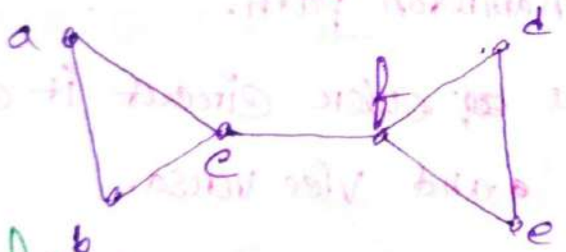
Pair (b, c) for:—

$deg(b)+deg(c) = 2+2 \geq n$
$= 4 \geq 4$
$=$ true.

(1) The Ore's theorem and Dirac's theorem provide sufficient conditions for a connected simple graph to have a Hamilton circuit.

problem: 37:—

⇒ Does the following graph have a Hamilton path? if so, find such the path. Does the above graph have Hamilton circuit.



⇒ For the following graph:—

→ It has a hamilton path namely a, b, c, f, d, e

→ The graph does not have hamilton circuit because it is not possible to start a vertex, and visit all the other vertex exactly one, then comback to the starting vertex without repeat the vertex.

Euler Vs. Hamilton

Paths and Circuits:-

(1) Euler path/circuit

⇒ All the EDGES must be visited exactly Once.

(2) Hamilton path/Circuit:-

⇒ All the Vertices must be visited exactly Once.

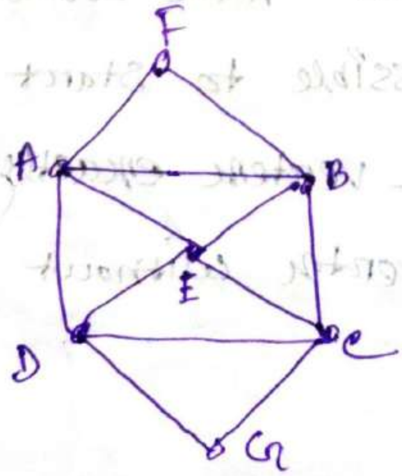
Note:-

(1) If a graph has a Hamilton circuit, then it Automatically has a hamilton path.

(2) If a graph has a euler circuit it cannot have a euler path and Vice versa.

Problem:- 1:-

⇒ Determine whether the following graph in a euler circuit and Hamilton circuit:



⇒ For the following graph:-

⇒ The graph has a euler circuit because all vertices has even degree.

One Such Euler Circuit is:-

A, F, B, E, D, C, E, A, D, G, C, B, A

⇒ The grap Also have hamilton circuit.

One Such Hamilton circuit is:-

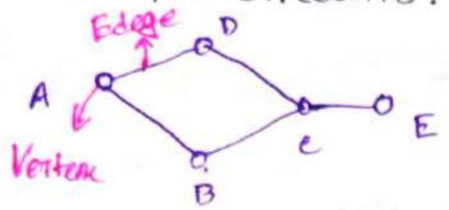
A, F, B, C, G, D, E, A.

Ans:-

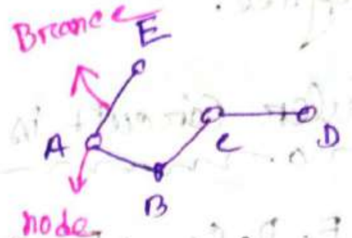
Lecture: 20:-

Tree:-

⇒ A tree is a connected undirected graph with no simple circuits.



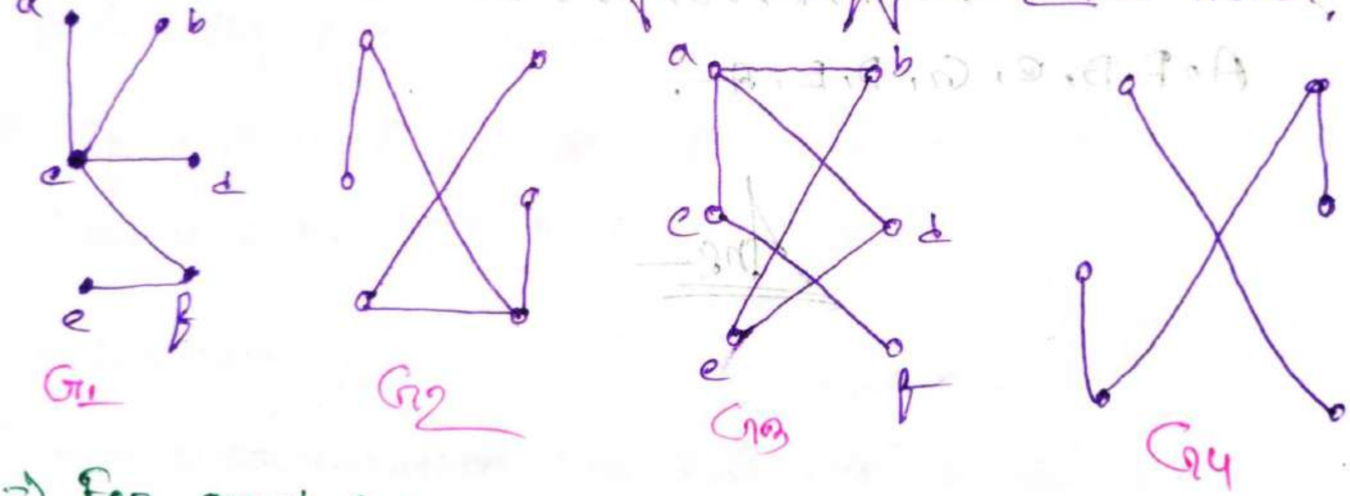
so, its a graph but not a tree.



so, its a graph and also a tree.

Problem: 1:-

⇒ Which of the graph showing in the figure 2 are tree?



⇒ For graph G1:-

→ The graph G1 is a tree because all the nodes are connected. And they dont have any simple circuit.

⇒ For graph: G2:-

→ Its also a tree. for the same reson of G1.

⇒ For G3:—

⇒ G3 is not a tree because of the graph contain a Simple circuit namely a, d, e, b, a.

⇒ For G4:—

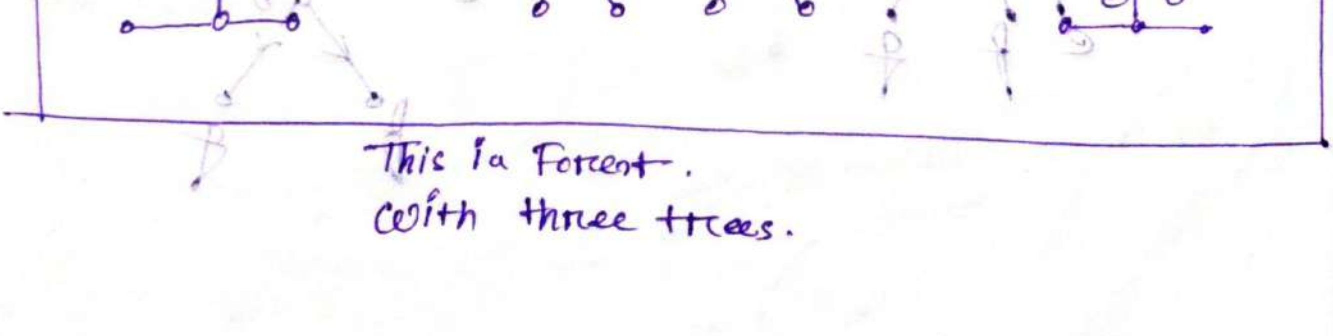
⇒ The graph is not a tree because it is not connected.

Note:— An Undirected graph is a tree iff there is a unique simple path between any two of its vertices.

Forest:

⇒ A forest is a graph that has no simple circuit, but is not connected. Each of the connected components in a forest is a tree.

Ex:—



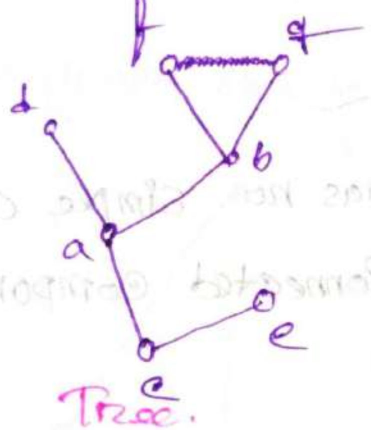
This is Forest.
with three trees.

Rooted Tree:-

⇒ A rooted tree is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

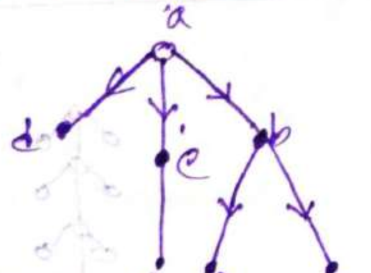
1. We can change an unrooted tree into a rooted tree by ~~shan~~ choosing any vertex as the root.
2. Different choies of the root produce diffrent rooted tree.

Ex:-

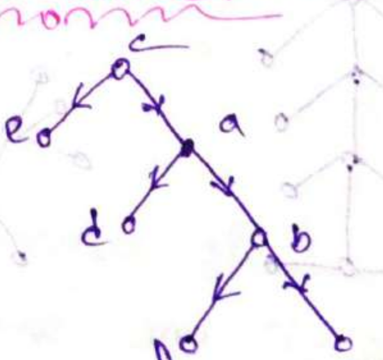


Tree.

With root a:-



With root c:-



Ordered Rooted Trees—

⇒ An ordered rooted tree is a tree where the children of each internal vertex are ordered.

(1.) We draw ordered rooted trees so that the children of each internal vertex are shown in order from left to right.

Binary Ordered Rooted trees:—

⇒ A Binary trees is an ordered rooted tree where each internal vertex has at most two children.

If an internal vertex of a binary tree has two children, the first is called the left child, and the second is called the right child.

The tree rooted at the left child of vertex is called the left subtree of the vertex, and the tree rooted at the right child of a vertex is called the right subtree of this vertex.

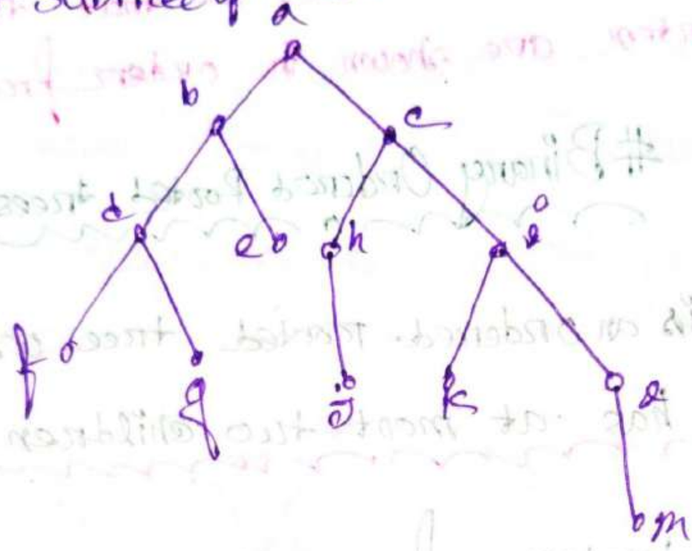
Problem: 4:-

⇒ Consider the binary tree T.

(i) What are the left and Right children of d?

(ii) what are the left subtree of c? and

(iii) right Subtree of c?



⇒

(i)

→ The left children of d is f

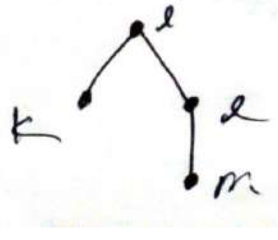
The Right children of d is g

(ii)

The left Subtree of c is:-

(iii)

The Right Subtree of c is:-



Rooted tree Terminologies:

Parent:

⇒ The node whic is predecessor of any node is called as parent node.

Child:

⇒ If u is parent of v, then V is called a child of u.

Siblings:

⇒ Vertices with the same parent are called siblings.

Leaf:

⇒ A Vertex of a rooted tree is called a leaf if it has no child.

Internal Vertex:

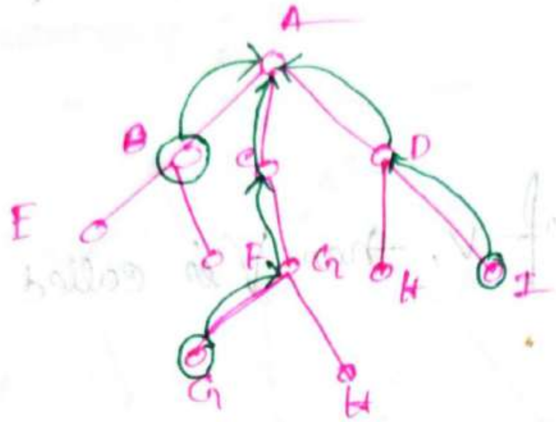
⇒ A Vertex that has a children is called internal Vertex.

Note:- The root is an internal vertex unless it is the only Vertex in the graph, in which case it is a leaf.

Ancestor(s):-

⇒ Any predessor nod on path from root to that note is refer to as an Ancestor.

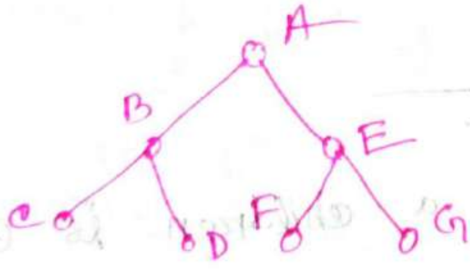
Ex:-



Desendant:-

⇒ Any successor node on any path from node to leaf node is refear to as an descendant.

Ex:-



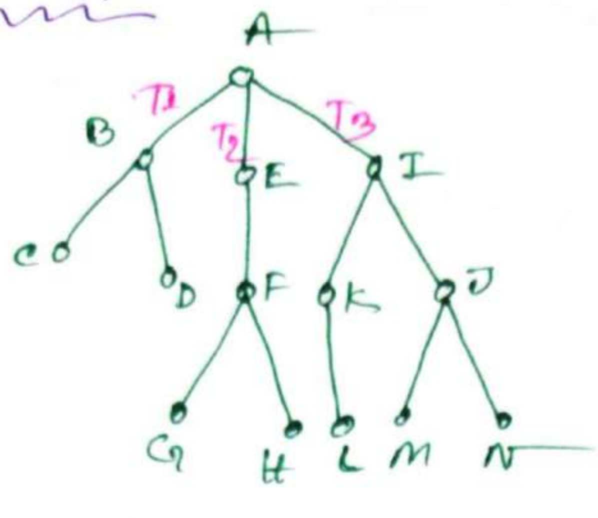
① Who are the desendant of Node B?

⇒ C, D are the denendant of Node B.

② Who are the desendant of root Node?

⇒ B, C, D, E, F, G are the desendant of Node A.

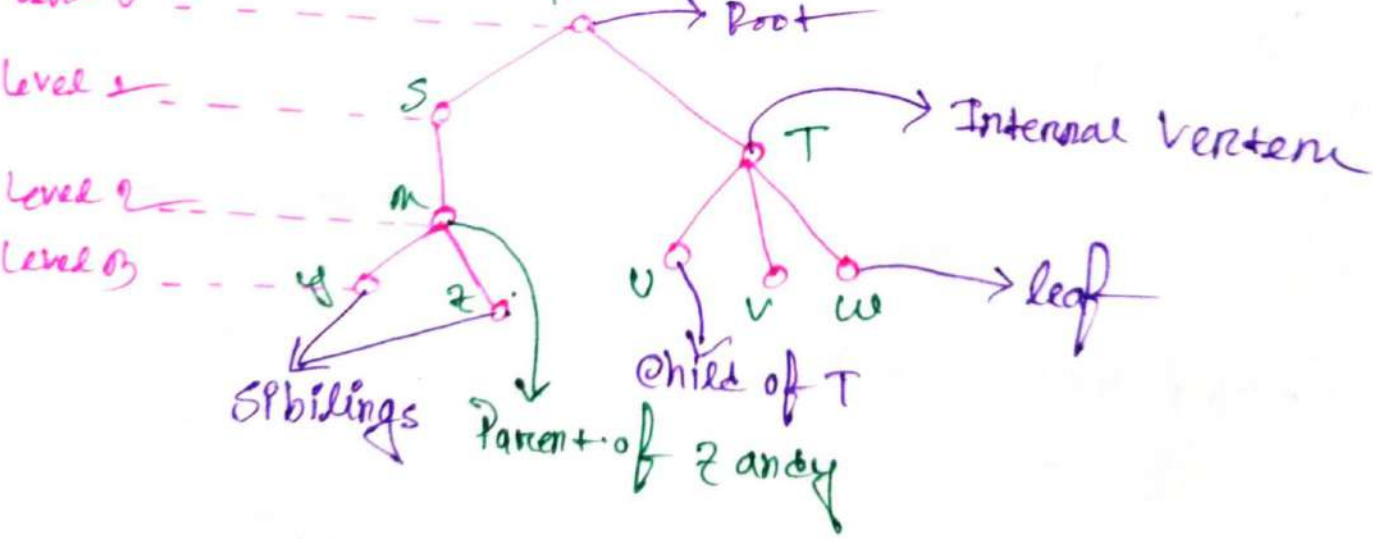
Sub-trees—



⇒ T_1, T_2, T_3 trees are the Subtree of A

⇒ Node with its descendants.

Illustration of Tree Terminology:



Tree as model:—

The family tree is a rooted tree where.

1. The Verticos represents family members
2. The edges represents a child relationship.

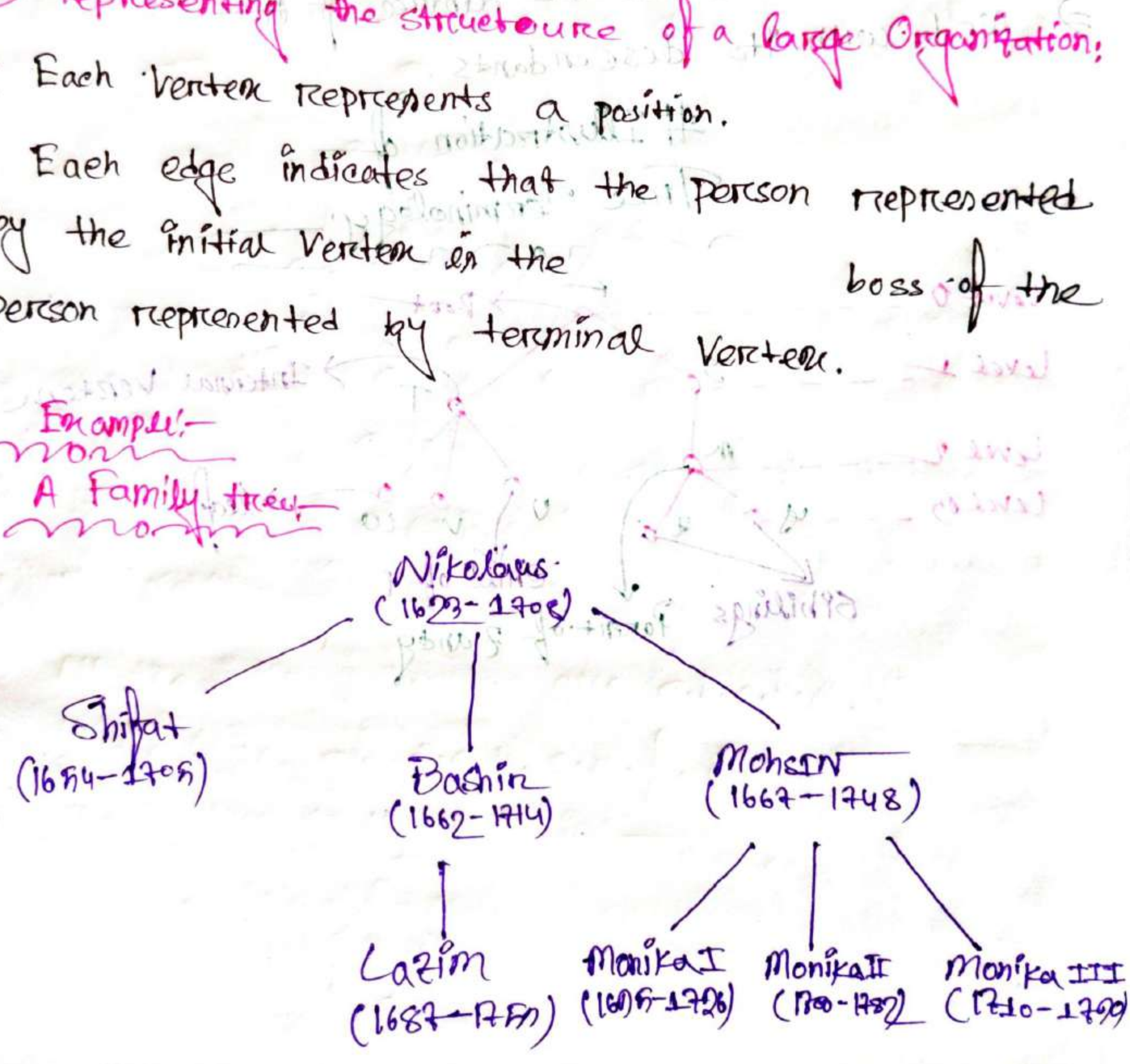
In business:—

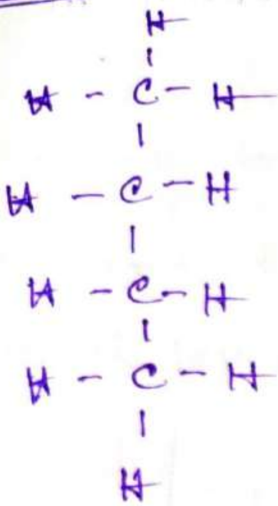
⇒ Representing the structure of a large Organization;

1. Each vertex represents a position.
2. Each edge indicates that the person represented by the initial Vertex is the boss of the person represented by terminal Vertex.

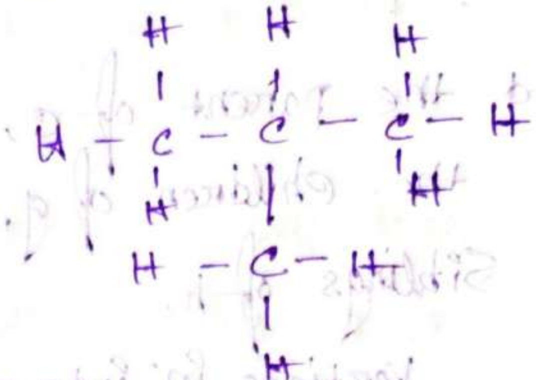
Example:—

A Family tree—



Ex:1 

Butane.

Ex:2 

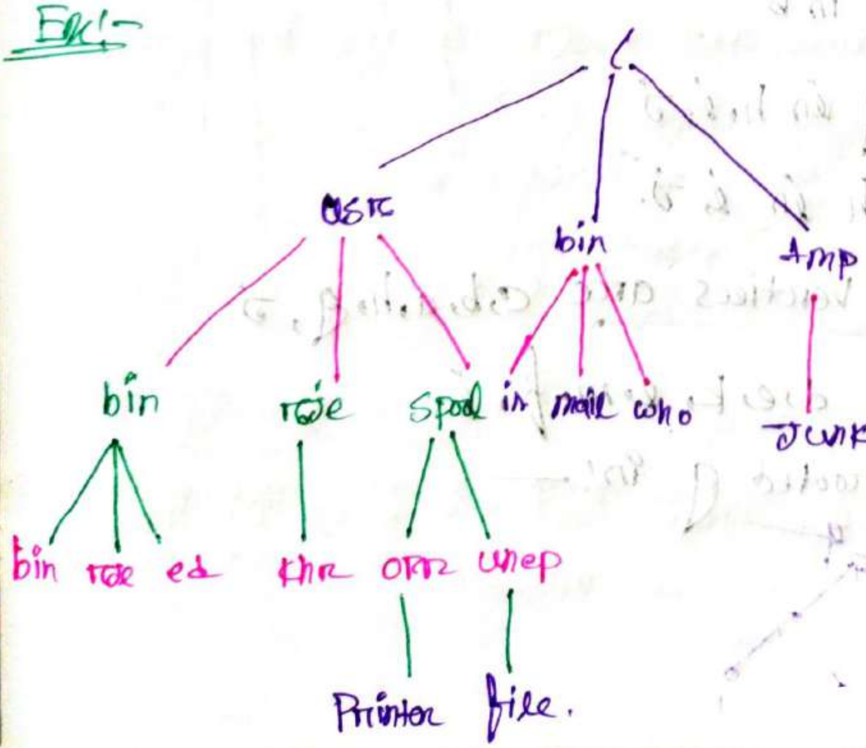
Isobutane.

In Computer Science:-

3) Representating Computer file systems:-

1. Each internal vertex represents a sub directory.
2. The root represents the root directory.
3. A leave represents an ordinary file or empty subdir.

Ex:-



Lecture: 24

In the rooted tree T (with root a) shown in the figure,

(i) Find the parent of c,

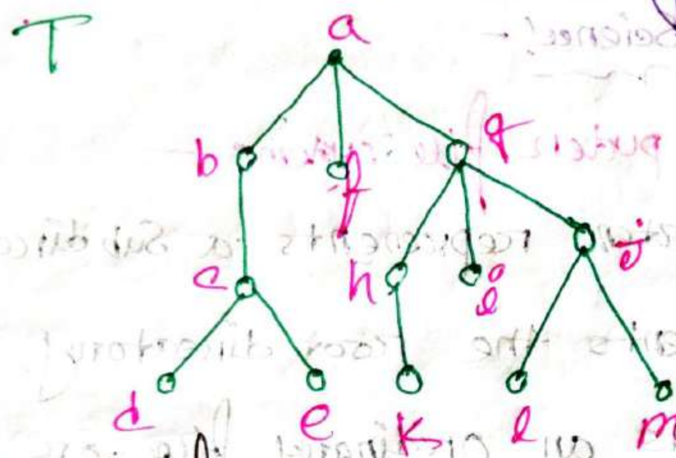
(ii) Find the children of g.

(iii) The siblings of h,

(iv) All the vertices in internal.

(v) All leaves,

(vi) what in the subtree rooted at g?



⇒ (i) The parent of c in b

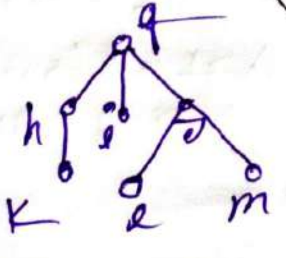
(ii) The children of g in h, i, j

(iii) The siblings of h in i, j.

(iv) All the internal vertices are c, b, a, h, g, j

(v) All the leaves are d, e, k, l, m, f, i

(vi) The subtree at rooted g in:—

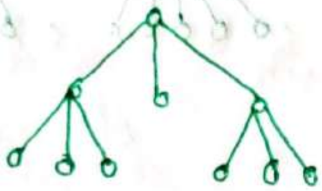


m-ary tree, full m-ary tree, Binary tree:—

m-ary tree:—

⇒ A rooted tree is called an m-ary tree if every internal vertex has no more than m children.

Ex:—

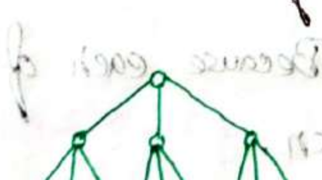


So, it's a 3-ary tree.

Full-mary tree:—

⇒ A rooted tree is called full-m-ary tree if every internal vertex has exactly m children.

Ex's



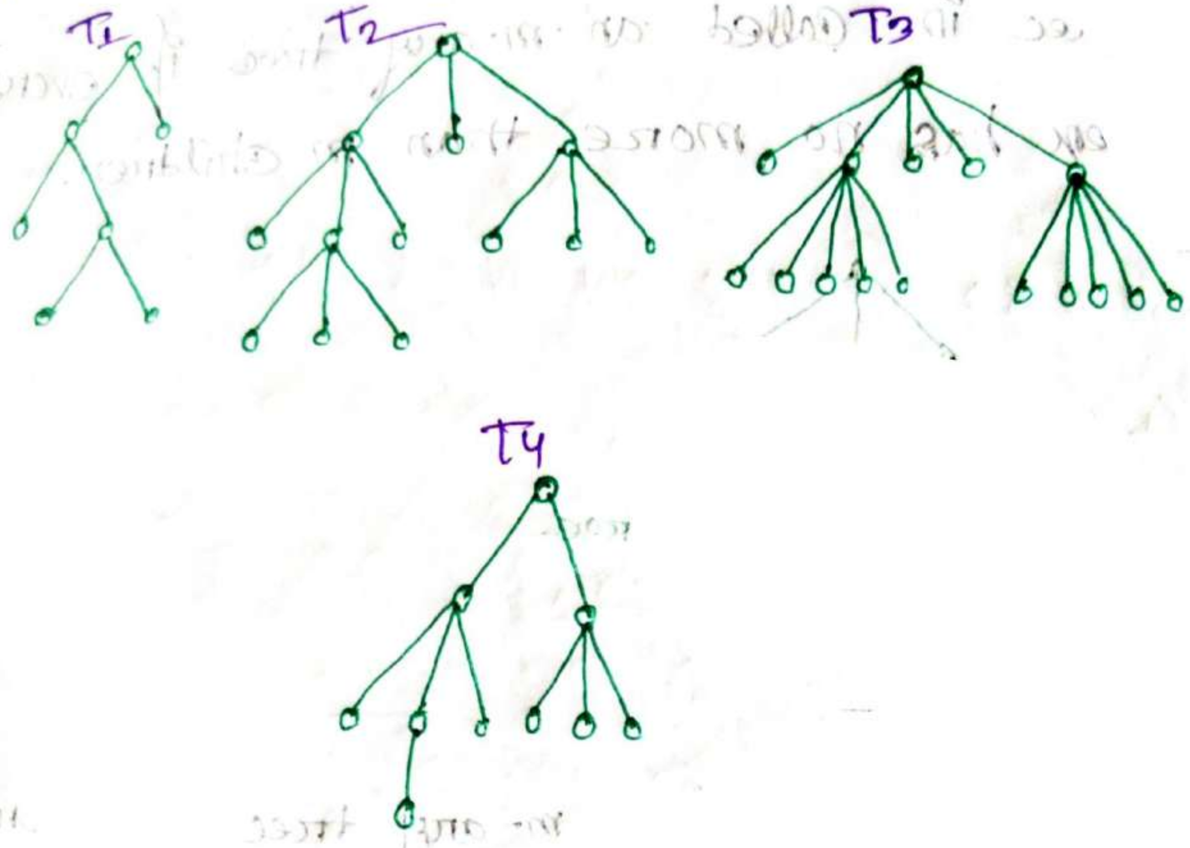
So, it's a full 3-ary tree.

Binary tree:—

⇒ An m-ary tree with $m = 2$ is called a binary tree.

Problem 3:-

⇒ Are the rooted trees in figure m-ary trees or full m-ary trees.



⇒ For tree T_1:-

→ T_1 is a full Binary tree Because each of its internal verties has two children.

⇒ For tree T_2:-

→ T_2 is a full 3-ary tree, Because each of its internal Verties has three children.

⇒ For tree T_3:-

→ T_3 is a full 5-ary tree, Because each of its internal vertices has 5/Five children.

⇒ For tree T_4:-

⇒ T_4 is not a full m-ary tree, for any m because some of its internal vertices have two children and others have three children.

Some Condition of m-ary tree:-

(1) A tree with n Vertices has $n-1$ edges.

(2) A full m-ary tree with i = internal vertices Contains $(n = mi+1)$ Vertices.

(3) For a full m-ary tree:-

(i) n-Vertices has $i = (n-1)/m$ → internal Vertices

$l = [(m-1)n+1]/m$ → leaves

(ii) i - internal Vertices has $n = mi+1$ → Vertices

$l = [(m-1)i+1]$ → leaves.

(iii) l - leaves has $n = (ml-1)/(m-1)$ → Vertices

$i = (l-1)/(m-1)$ → Internal vertices.

(4) There are at most $= m^h$ leaves in an m-ary tree

h = height

Problem: 17

⇒ How many edges does a tree with 10,000 Vertices have?

⇒ In given:

$n = 10,000$

We know:

A tree with n-Vertices with has $n-1$ edges

$= (10,000 - 1)$ edges.
$= 9999$ edges.

Ans

Problem: 18

⇒ How many Vertices does a full 5-ary tree with 100 internal vertices have?

⇒ In given:

$m = 5$
$i = 100$

We know:

$n = mi + 1$
$= 100 \times 5 + 1$
$= 501$ Vertices.

$l = (m-1) i + 1$
$= (100-1) 5 + 1$
$= 496$ leaves.

Problem: 101 —

⇒ How many Edges does a full binary tree with 100 internal vertices have?

⇒ In given:—

$m = 2$

$i = 100$

we know —

$n = (mi + 1)$

$= 100 \times 2 + 1$

$= 201$ Vertices

Edges $= n - 1$

$= 201 - 1$

$= 200$ Edges.

How many leaves does a full 3-ary tree with 100 Vertices have?

⇒ In given —

$m = 3$

$n = 100$

we know: —

$l = [(m-1) \times n + 1]/m$

$= (3-1) \times 100 + 1/3$

$= 67$

$l = [(m-1) \times n + 1]/m$

$= (2) \times 100 + 1/3$

$= 67$ leaves

$i = (n-1)/m$

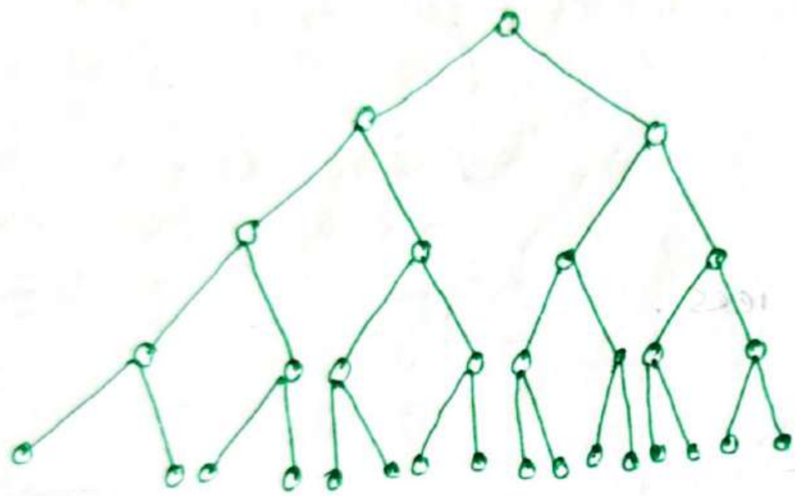
$= 99/3$

$= 33$ internal Vertices.

Problem: 27:-

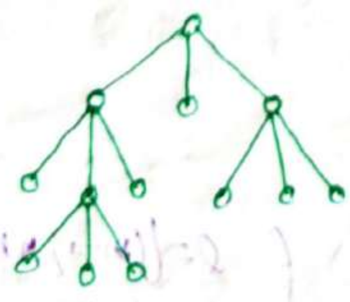
⇒ Construct a complete binary tree of height 4 and complete 3-ary tree of height 3.

⇒ Complete Binary tree:-



It's a complete Binary tree with height 4

Complete 3-ary tree of height 3:-



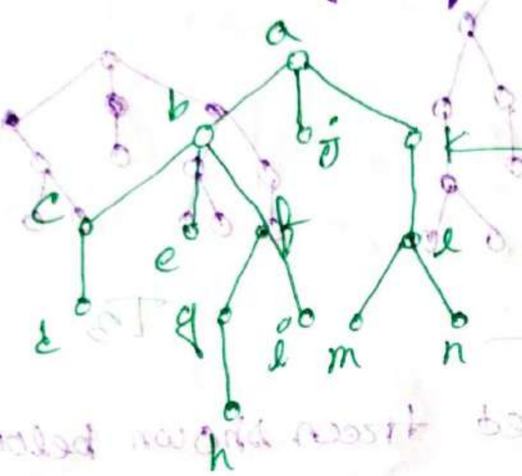
It's a complete 3-ary tree with height 4

Level of Vertices and Height of trees :-

Ex:-

(i) Find the level of Each Vertex in the tree.

(ii) What is the height of the tree.



Ans (i) ① The root a is at level 0.

(ii) Vertices b, j, k at level 1

(iii) Vertices c, e, f, l at level 2

(iv) Verties d, g, i, m, n, at level 3

(v) Vertex h at level 4

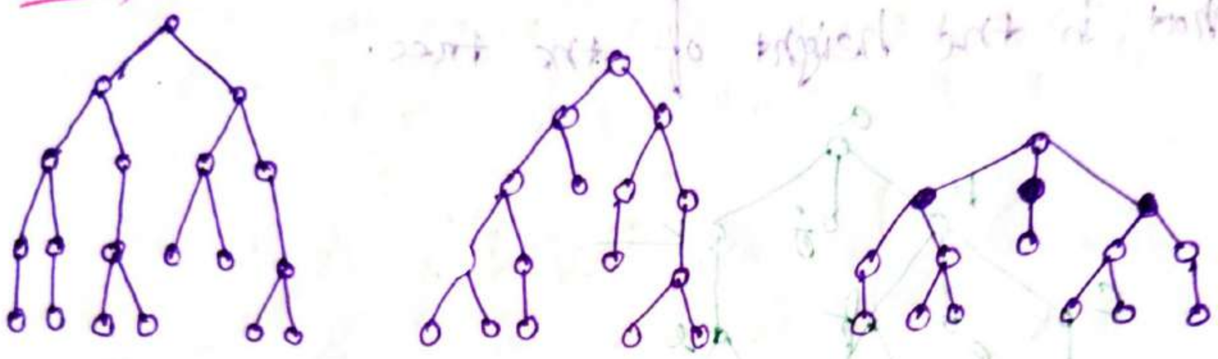
(ii) The height of the tree = 4

Since 4 is the largest level of the vertices in this tree.

Balanced m-ary tree:-

⇒ A rooted m-ary tree of height h is blanced if all leaves are at levels h or h-1.

Exe:-



T_1 T_2 T_3

⇒ Which of the rooted treen shown below is blanced?

⇒ For T_1 and T_3:-

⇒ Those two tree are blanced m-ary tree because T_1 has all the leaves in level 4 and level 3 and T_2 han all the leaves in 3 and level 2.

⇒ For T_2:-

⇒ Highest number of level = 4
· T_2 tree are not blanced m-ary tree because T_2 han leaves in level 4, 2, 3.

Lecture 10:—

Primes and Composite Number:—

Prime Number:—

⇒ A positive integer p greater than 1 is called prime number if the only positive factors of p are 1 and

Composite Number:—

⇒ A positive integer that is greater than ~~1~~ and is not prime is called composite.

Note:—

⇒ The integer n is composite iff there exists an integer a such that $a|n$ and $1<a<n$.

(1) There are infinitely many primes.

Problem: 1:—

⇒ Why the integer 7 is prime and why the integer 9 is composite.

⇒ Integer 7 is prime because it's only positive factor are 1, 7.

⇒ The integer 9 is a composite because it is divisible by three/3.

lecture 10:—

Primes and Composite Number:—

Prime Number:—

⇒ A positive integer p greater than 1 is called prime number if the only positive factors of p are 1 and p

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⇒ A positive integer that is greater than 1 and is not prime is called composite.

Note:—

⇒ The integer n is composite iff there exists an integer a such that $a \mid n$ and $1 < a < n$.

(1) There are infinitely many primes.

Problem: 1:—

⇒ Why the integer 7 is prime and why the integer 9 is composite.

⇒ Integer 7 is prime because it's only positive factor are 1, 7.

⇒ The integer 9 is a composite because it is divisible by three/3.

Problem: 2:-

⇒ What are the primes less than 100?

⇒ There is:-

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 89, 83, 97.

Fundamental Theorem of Arithmetic:-

⇒ Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes, where the prime factor are written in order of non-decreasing size.

Problem: 2:-

⇒ What are the prime factorization of 100, 641, 999 and 1024.

⇒ $100 = 2\cdot2\cdot5\cdot5 = 2^2\cdot5^2$

$641 = 641$

$999 = 3\cdot3\cdot3\cdot37 = 3^3\cdot37$

$1024 = 2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2 = 2^{10}$

Determining whether a given integer is prime or Composite:-

⇒ If n is a composite integer, then n has a prime division less than or equal to $\sqrt{n}$.

Ex:- Let, $n = 10$

$\therefore \sqrt{n} = \sqrt{10} = 3.16$

$a|n = 2|10 = 5$

⇒ If n is a prime integer, then n is not divisible by any prime number less than or equal to its squre root.

Problem:-

⇒ Determine whether a given integer is prime or Composite-

(1.) Show that 101 is prime.

⇒ In given:-

$n = 101$

$\therefore \sqrt{101} = 10.04$

The only prime number not greater than $\sqrt{101}$ is 2, 3, 7, 5. Since 101 not divisible by 2, 3, 5, 7 so, its a prime number.

(2.) Determine 111 is prime:-

⇒ In given:- $n = 111$

The only prime number not greater than $\sqrt{111}$ is 2, 3, 5, 7. Since 111 is divisible by 3 so, its not a prime number.

(3) Determine whether 143 is prime?

⇒ In given—

$n = 143$

The only prime number not greater than $\sqrt{143}$ are 2, 3, 5, 7, 11.

Since 143 is divisible by 11, so, its not a prime number.

(4) 139 is prime or not?

⇒ In given—

$n = 139$

The only prime number not greater than $\sqrt{139}$ are 2, 3, 5, 7, 11.

Since 139 is not divisible by 2, 3, 5, 7, 11

So, it's a prime number.

Greatest Common Divisor (gcd)

⇒ Let a and b be integer, not both zero. The largest integer d such that d|a and d|b is called the gcd of a and b, denote by gcd(a,b)

Problem: 19—

⇒ What are the gcd of 24 and 36.

⇒ The positive common divisor of 24 · 36 are

1, 2, 3, 4, 6, 12

∴ gcd(24, 36) = 12

Relatively prime